ANSWER KEY Reduced copy of the question page, with its answer

3 of 200 runs is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

2401

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

6 + 18 runs out of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

2402

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

19 out of 200 is 19%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

2403

QUESTION ID

S64-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 19/200 = 0.095 = 9.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

34 of 200 is

A. 1.7%

B. 17%

C. 6.8%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

2404

QUESTION ID

S64-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17%

GRADING NOTE 34/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 2% less than 5%? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

2405

QUESTION ID

S64-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Compare the percents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 1

Design a simulation to test a model • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A player claims to make 70% of her free throws, and she made 11 of 20. Which simulation tests the claim?

A. Digits 0-6 = make; 20 digits per run; repeat many runs; find how often 11 or fewer makes occur.

B. Digits 0-6 = make; 20 digits in a single run; the claim is true if that run has 14 makes.

C. Digits 0-4 = make; 20 digits per run; repeat many runs; find how often exactly 11 makes occur.

D. Digits 0-6 = make; 11 digits per run; repeat many runs; find how often 20 makes occur.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

2407

QUESTION ID

S64-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Digits 0-6 = make; 20 digits per run; repeat many runs; find how often 11 or fewer makes occur.

GRADING NOTE 7 of 10 digits model 70%; each run copies 20 shots; count results at least as extreme.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A player who claims to make 70% of free throws made 11 of 20. Why should a simulation count runs with 11 or fewer makes, not only runs with exactly 11?

A. Any one exact count can be rare; the question is how often results at least this far below the claim occur.

B. Runs with exactly 11 makes are impossible under the model, so they would never be counted at all.

C. The claim is about making 11 or fewer shots, not about making 70% of her free throws.

D. Counting 11 or fewer always makes the result significant, which is the goal of the simulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

2408

QUESTION ID

S64-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Any one exact count can be rare; the question is how often results at least this far below the claim occur.

GRADING NOTE Use results as extreme or more extreme.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

True or false?

A result that happens in 30% of simulated runs is strong evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

2409

QUESTION ID
S64-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 30% is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A common cutoff for "unusual" is

A. about 50% of runs

B. 100% of runs

C. about 5% of runs

D. about 25% of runs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

2410

QUESTION ID

S64-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. about 5% of runs

GRADING NOTE Convention.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

A simulation of a model is run 400 times instead of 200 times. What does this change?

A. It changes the observed data that the model is compared with.

B. It changes the probabilities that the model assigns to each outcome.

C. It turns an unusual result into a usual one, because more runs are counted.

D. It gives a more precise estimate of how often a result occurs under the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

2411

QUESTION ID

S64-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. It gives a more precise estimate of how often a result occurs under the model.

GRADING NOTE More runs estimate the model's proportion more precisely; the model and data are unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

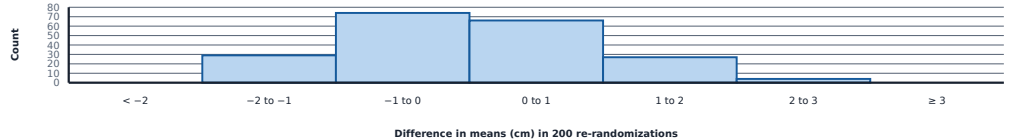
SET 64 • HSS-IC.A.2

FORWARD 2

Compare two treatments with a re-randomization simulation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, 20 plants were randomly assigned to fertilizer or no fertilizer, and the fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences under the model that fertilizer has no effect; none was 2.9 cm or more. Use 5% as the cutoff for unusual. What do you conclude?



A. 0/200 < 5%: a difference this large is very unlikely by chance, so fertilizer appears to increase growth.

B. The difference is not significant, because no re-randomized difference matched it exactly.

C. 0/200: a difference this large has probability exactly 0, so the fertilizer is proved to work.

D. No conclusion is possible, because the result must first be compared with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

2413

QUESTION ID

S64-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 0/200 < 5%: a difference this large is very unlikely by chance, so fertilizer appears to increase growth.

GRADING NOTE A rare result is evidence of an effect, not proof; probability is not shown to be zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

What model does a re-randomization simulation assume?

A. The two groups were chosen by voluntary response.

B. The observed difference is exactly the true effect.

C. The treatment has no effect, so the group labels can be shuffled.

D. The treatment has the same large effect on every subject.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

2414

QUESTION ID

S64-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The treatment has no effect, so the group labels can be shuffled.

GRADING NOTE The null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

True or false?

Re-randomization is a simulation of a model, like the coin-flip simulation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

2415

QUESTION ID

S64-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both test a model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Under the model of no treatment effect, 60 of 200 re-randomized differences were at least as large as the observed one. Use 5% as the cutoff for unusual. The result is

A. significant

B. proof of no effect

C. not statistically significant

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

2416

QUESTION ID

S64-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. not statistically significant

GRADING NOTE 30% is common; not significant does not prove the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

How does a re-randomization test for an experiment extend the reasoning of testing a fair-coin claim with a simulation?

A. Both flip a real coin many times and then count how many heads appear in each of the groups of subjects in the study.

B. Both ask how often a model produces a result at least as extreme as the observed one; here the model is "no treatment effect."

C. The coin test uses a probability model, but the re-randomization test does not use any model of the results at all.

D. Both prove the model is true whenever the observed result turns out to be common among the simulated results.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

2417

QUESTION ID

S64-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Both ask how often a model produces a result at least as extreme as the observed one; here the model is "no treatment effect."

GRADING NOTE Same logic, different null model. The others describe flipping a real coin, deny that re-randomization uses a model, or treat a common result as proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Identify sample surveys, experiments, and observational studies

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Statistical questions and data collection	**6.SP.A.1**
BACKWARD 2	Association is not causation	**HSS-ID.C.9**
BACKWARD 3	Random samples	**7.SP.A.1**
FORWARD 1	Identify confounding and design flaws	**HSS-IC.B.3**
FORWARD 2	Analyze the results of an experiment	**HSS-IC.B.5**

SET 65 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Researchers randomly assign 100 volunteers to take a vitamin or a placebo and compare the number of colds. What type of study is this, and what is its purpose?

A. Experiment: to estimate the percent of all people who catch colds.

B. Observational study: to describe colds without any treatment.

C. Experiment: to test whether the vitamin causes fewer colds.

D. Sample survey: to estimate the percent of people who take vitamins.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

2420

QUESTION ID
S65-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Experiment: to test whether the vitamin causes fewer colds.

GRADING NOTE Random assignment of a treatment makes it an experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

A pollster asks a random sample of 1,000 voters which candidate they prefer. What type of study is this, and what is its purpose?

A. Sample survey: to estimate the preference of all voters.

B. Experiment: to test whether a candidate causes votes.

C. Observational study: to show which candidate causes turnout.

D. Sample survey: to show the poll caused voters' choices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

2421

QUESTION ID
S65-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Sample survey: to estimate the preference of all voters.

GRADING NOTE A random sample survey estimates a population value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

Researchers compare the hospital records of smokers and nonsmokers for lung disease. What type of study is this, and can it by itself show causation?

A. Experiment; yes, smoking was the treatment in the study.

B. Observational study; no, the groups were not randomly assigned.

C. Observational study; yes, hospital records are accurate.

D. Sample survey; yes, the sample of records is large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

2422

QUESTION ID

S65-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Observational study; no, the groups were not randomly assigned.

GRADING NOTE No treatments were assigned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

Which study best answers "Does playing music while studying improve test scores?"

A. a census of how many students in the school own headphones

B. an observational study of students who already choose to play music

C. a survey that asks students whether they think music helps them study

D. an experiment: randomly assign students to study with or without music and compare scores

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

2423

QUESTION ID
S65-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. an experiment: randomly assign students to study with or without music and compare scores

GRADING NOTE A cause-and-effect question needs an experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

Which study best answers "What percent of students at the school use the library weekly?"

A. a sample survey of a random sample of the school's students

B. a voluntary online poll posted on the library website

C. a randomized experiment that assigns students to use the library

D. an observational study of students in the library today

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

2424

QUESTION ID

S65-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. a sample survey of a random sample of the school's students

GRADING NOTE Estimating a percent calls for a random sample survey.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 1

Statistical questions and data collection • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How many hours do students at our school sleep?" a statistical question?

A. No: it has a single correct answer.

B. Yes: the answers vary from student to student.

C. No: hours of sleep cannot be measured.

D. Yes: it asks about one particular student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

2426

QUESTION ID

S65-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Yes: the answers vary from student to student.

GRADING NOTE A statistical question anticipates variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which method of collecting data best answers "What percent of the city's voters favor the new library?"

A. an experiment that assigns voters to read books

B. a survey of people leaving the old library

C. asking the members of the city council

D. a survey of a random sample of the city's voters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

2427

QUESTION ID

S65-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. a survey of a random sample of the city's voters

GRADING NOTE Estimating a population percent calls for a random sample survey.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An experiment imposes a treatment on the subjects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

2428

QUESTION ID

S65-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

Watching and recording without imposing treatments is

A. an experiment

B. an observational study

C. a census

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

2429

QUESTION ID

S65-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. an observational study

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

In an experiment, what is a placebo?

A. a random sample of the population

B. the group that gets the real treatment

C. a fake treatment with no active ingredient

D. a measurement taken before the study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

2430

QUESTION ID

S65-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a fake treatment with no active ingredient

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 2

Association is not causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Ice cream sales and sunburns are positively correlated. Which explanation is best supported?

A. Sunburns cause ice cream sales, because a correlation shows the direction of cause.

B. The correlation must be a calculation error, because the variables are unrelated.

C. Ice cream causes sunburn, because the two variables are positively correlated.

D. Hot, sunny weather increases both; the correlation does not show that ice cream causes sunburn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

2432

QUESTION ID

S65-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Hot, sunny weather increases both; the correlation does not show that ice cream causes sunburn.

GRADING NOTE A lurking variable explains the association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

"Students with more books at home get higher test scores." Which is a plausible lurking variable?

A. family income

B. the test's page count

C. the student's ID number

D. the color of the book covers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

2433

QUESTION ID

S65-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. family income

GRADING NOTE Income can affect both books at home and scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

True or false?

A strong correlation proves that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q3/5

2434

QUESTION ID

S65-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Correlation is not causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q3/5

A variable that affects both variables in a study is

A. the placebo

B. the sample

C. a confounding (lurking) variable

D. the response

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

2435

QUESTION ID

S65-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a confounding (lurking) variable

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which kind of study can show cause and effect?

A. a voluntary-response survey

B. an observational study

C. a randomized experiment

D. a sample survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

2436

QUESTION ID

S65-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 3

Random samples • Grade 7 • Statistics & Probability • 7.SP.A.1

Which method gives a simple random sample of 50 students from a school list?

A. Choose 50 students who are standing in the cafeteria line at lunch.

B. Number the students and use a random number generator to pick 50 different numbers.

C. Ask for 50 volunteers by posting a sign-up sheet in the school office.

D. Choose the first 50 names on the list, which is in alphabetical order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

2438

QUESTION ID

S65-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Number the students and use a random number generator to pick 50 different numbers.

GRADING NOTE Every group of 50 has the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

Is surveying the students in the cafeteria line a random sample of the school?

A. Yes: the students in the line arrived in a random order.

B. No: a random sample must include every student.

C. No: students who are not in the line have no chance to be chosen.

D. Yes: any student could be in the cafeteria line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

2439

QUESTION ID

S65-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. No: students who are not in the line have no chance to be chosen.

GRADING NOTE Convenience sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A random sample tends to represent the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

2440

QUESTION ID

S65-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No systematic bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is random?

A. the teacher's friends among the students

B. students who volunteer when asked in class

C. names drawn from a hat with every student's name

D. the first 50 students to arrive at school

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

2441

QUESTION ID

S65-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. names drawn from a hat with every student's name

GRADING NOTE Chance selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does a survey need a random sample?

A. so it needs to ask only 2 people

B. so every person gives the same answer

C. so it can show cause and effect by itself

D. so its results can be generalized to the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

2442

QUESTION ID

S65-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. so its results can be generalized to the population

GRADING NOTE Generalization. The others are false: random samples are not limited to 2 people, do not make answers equal, and do not show cause and effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 1

Identify confounding and design flaws • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Students who choose to take an SAT prep course score higher than students who do not. Which is a confounding variable, with a correct reason?

A. Hair color: it differs between students, so it must confound the result of the study.

B. Motivation: students who choose the course may be more motivated, which also raises scores.

C. The test date: all students take the same test, so the date explains the difference.

D. The course itself: it raises scores, so it is the confounding variable in the study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

2444

QUESTION ID

S65-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Motivation: students who choose the course may be more motivated, which also raises scores.

GRADING NOTE A confounder is related to both the choice and the outcome.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Students who choose to take an SAT prep course score higher than students who do not. Which design could show whether the course causes higher scores?

A. Randomly assign volunteers to take the course or not, then compare their scores.

B. Let students choose whether to take the course again, but use a larger sample.

C. Compare the scores of students at two different schools that offer the course.

D. Survey many more students who chose to take the course about their scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

2445

QUESTION ID
S65-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Randomly assign volunteers to take the course or not, then compare their scores.

GRADING NOTE Random assignment breaks the link with confounders.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

True or false?

An experiment without a control group can still show the treatment caused the result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

2446

QUESTION ID

S65-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Comparison is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

In an experiment, why use a placebo?

A. to make the study observational instead

B. to control for believing one is treated

C. to increase the sample size of the study

D. to make the real treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

2447

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. to control for believing one is treated

GRADING NOTE Placebo effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

Why is an observational study, rather than an experiment, used to study whether smoking causes lung disease in people?

A. Experiments cannot use random assignment.

B. It would be unethical to assign people to smoke.

C. Smokers cannot be counted in an experiment.

D. Observational studies always show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

2448

QUESTION ID

S65-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. It would be unethical to assign people to smoke.

GRADING NOTE Ethics. The others are false: experiments do use random assignment, smokers can be counted, and observational studies do not show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 2

Analyze the results of an experiment • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

100 volunteers were randomly assigned to a vitamin or a placebo; the vitamin group had 1.2 fewer colds on average. Under the model that the vitamin has no effect, 40 of 200 re-randomizations gave the vitamin group at least 1.2 fewer colds. Use 5% as the cutoff for unusual. What do you conclude?

A. 40/200 = 20% > 5%: the difference could easily be chance, so it is not significant.

B. 40/200 = 20% > 5%: this proves the vitamin has no effect.

C. 40/200 = 20%: the vitamin prevents 20% of colds in these volunteers.

D. 40/200 = 20% > 5%: the difference is significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

2450

QUESTION ID

S65-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A.
40/200 = 20% > 5%: the difference could easily be chance, so it is not significant.

GRADING NOTE Not significant is not proof of no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized vitamin-versus-placebo experiment, the vitamin group had 1.2 fewer colds on average. Under the model that the vitamin has no effect, 2 of 200 re-randomized differences were that large in the vitamin's favor. Use 5% as the cutoff for unusual. What do you conclude?

A. 2/200 = 1%: this proves the vitamin works for everyone.

B. 2/200 = 1%: the vitamin works for only 1% of people.

C. 2/200 = 1% < 5%: significant, so the vitamin appears to reduce colds.

D. 2/200 = 1% < 5%: the difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

2451

QUESTION ID

S65-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 2/200 = 1% < 5%: significant, so the vitamin appears to reduce colds.

GRADING NOTE A significant result is evidence, not proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

True or false?

Random assignment makes it possible to attribute a significant difference to the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

2452

QUESTION ID
S65-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Groups differ only by chance otherwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The re-randomization model assumes

A. the groups are different sizes

B. the treatment has no effect

C. the treatment works

D. the data are normal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

2453

QUESTION ID

S65-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the treatment has no effect

GRADING NOTE Null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

Why can a significant difference in an observational study not be interpreted as a treatment effect the way it can in a randomized experiment?

A. Observational studies always use smaller samples, so their differences are always due to chance.

B. Observational studies cannot measure any differences between groups, so no difference can be found.

C. A significant difference in an observational study is always the result of a calculation error.

D. Without random assignment, the difference could come from confounding variables rather than the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2454

QUESTION ID

S65-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Without random assignment, the difference could come from confounding variables rather than the treatment.

GRADING NOTE Confounding. The others make false claims about sample size, measurement, and calculation errors; the real issue is confounding.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

SET 66 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Explain random sampling versus random assignment

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Study types and their purposes	**HSS-IC.B.3**
BACKWARD 2	Choose a random sample	**7.SP.A.1**
BACKWARD 3	Correlation and causation	**HSS-ID.C.9**
FORWARD 1	Improve an experimental design	**HSS-IC.B.3**
FORWARD 2	Use re-randomization to judge a treatment difference	**HSS-IC.B.5**

SET 66 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 500 adults is surveyed about exercise and sleep; adults who exercise more tend to sleep better. Which conclusion is justified?

A. Exercise causes better sleep for all adults, because the sample was chosen at random.

B. The association can be generalized to all adults, but it does not show that exercise causes better sleep.

C. Nothing can be concluded about any adults beyond the 500 who were surveyed.

D. Exercise causes better sleep, but only for the 500 adults who were surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

2457

QUESTION ID
S66-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. The association can be generalized to all adults, but it does not show that exercise causes better sleep.

GRADING NOTE Random sample: generalize. No random assignment: no causation. The others claim causation from a survey, limit the association to the 500 surveyed, or deny any conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers are randomly assigned to two diets, and diet A's group loses significantly more weight. Which conclusion is justified?

A. There is evidence that diet A caused more weight loss for these volunteers, but the result cannot be generalized to all people.

B. Diet A is associated with weight loss for all people, but the study cannot show that the diet caused it.

C. Diet A causes more weight loss for all people, because the volunteers were randomly assigned to the two diets.

D. Nothing can be concluded, because the volunteers chose to participate instead of being randomly sampled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

2458

QUESTION ID
S66-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. There is evidence that diet A caused more weight loss for these volunteers, but the result cannot be generalized to all people.

GRADING NOTE Random assignment: evidence of cause for these subjects. No random sample: no generalization. The others generalize to all people, deny causation despite random assignment, or deny any conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

A random sample of 200 students from a school is randomly assigned to two study methods, and method 1 leads to significantly higher scores. Which conclusion is justified?

A. Method 1 is associated with higher scores, but the study cannot show that it causes them.

B. There is evidence that method 1 causes higher scores, but only for the 200 students in the study.

C. There is evidence that method 1 causes higher scores, and the result can be generalized to the school's students.

D. Method 1 causes higher scores for students at every school, because the result was significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

2459

QUESTION ID
S66-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. There is evidence that method 1 causes higher scores, and the result can be generalized to the school's students.

GRADING NOTE Random sample and random assignment: both conclusions, limited to this school. The others limit the result to the 200 students, deny causation despite random assignment, or generalize beyond the school.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A teacher compares test scores of students in her class who chose to use a review app with those who did not. Which conclusion is justified?

A. The app causes higher scores, but only for the students in this class.

B. The app causes higher scores for all students, because the groups were compared.

C. The result can be generalized to all students, but it does not show causation.

D. Only an association in this class: there was no random assignment and no random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

2460

QUESTION ID
S66-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. Only an association in this class: there was no random assignment and no random sample.

GRADING NOTE Neither random assignment nor random sampling. The others claim causation or generalization, which need random assignment and random sampling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

How can 40 subjects be randomly assigned to two groups of 20?

A. Put the first 20 subjects who arrive in group 1 and the other 20 subjects in group 2.

B. Put the 20 tallest subjects in group 1 and the 20 shortest subjects in group 2.

C. Let each subject choose a group, and stop when one of the groups has 20 subjects.

D. Number the subjects 1-40; choose 20 different random numbers for group 1; the rest form group 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

2461

QUESTION ID

S66-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Number the subjects 1-40; choose 20 different random numbers for group 1; the rest form group 2.

GRADING NOTE Chance alone decides the groups. The others let arrival order, choice, or height decide the groups, so the groups may differ systematically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 1

Study types and their purposes • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Which type of study uses random assignment of treatments?

A. an observational study

B. an experiment

C. a sample survey

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

2463

QUESTION ID

S66-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. an experiment

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

Which type of study is designed to estimate a population value?

A. a case study of one person

B. an experiment

C. a sample survey

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

2464

QUESTION ID

S66-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. a sample survey

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An observational study assigns treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

2465

QUESTION ID

S66-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It observes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Random assignment is used to

A. avoid measuring

B. make the sample larger

C. make the groups comparable

D. choose who is in the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

2466

QUESTION ID

S66-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. make the groups comparable

GRADING NOTE Comparability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

What does random sampling allow you to do?

A. generalize from the sample to the population

B. conclude cause and effect from the data

C. remove all chance variation from results

D. skip measuring the people in the sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

2467

QUESTION ID

S66-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. generalize from the sample to the population

GRADING NOTE Random sampling supports generalization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 2

Choose a random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

Which method chooses 20 names at random from a numbered list of 500?

A. Pick 20 different random numbers from 1 to 500 and take those names.

B. Pick 20 names that look like a good mix of the whole list.

C. Take the first 20 names on the numbered list, numbers 1 to 20.

D. Email all 500 and take the first 20 people who reply to the email.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

2469

QUESTION ID

S66-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Pick 20 different random numbers from 1 to 500 and take those names.

GRADING NOTE Random numbers give every name the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is a sample of volunteers a random sample? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

2470

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sample of volunteers may differ from the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

2471

QUESTION ID

S66-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Self-selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

Which method gives a random sample?

A. choosing the first 20 people on the list

B. drawing numbered slips from a well-mixed hat

C. asking for volunteers to sign up

D. asking friends to take part

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

2472

QUESTION ID

S66-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. drawing numbered slips from a well-mixed hat

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

Names are written on slips and drawn from a hat. Why must the slips be well mixed?

A. so the drawing takes longer

B. so the first slip drawn is always the same

C. so every slip has the same chance of being drawn

D. so the hat can hold more slips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

2473

QUESTION ID

S66-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. so every slip has the same chance of being drawn

GRADING NOTE Equal chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 3

Correlation and causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Can an association between two variables in observational data prove causation? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

2475

QUESTION ID

S66-B3-Q1

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Students who exercise more tend to sleep better. Which is a plausible lurking variable?

A. the day of the month they were born

B. stress level

C. their locker number

D. the brand of their shoes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

2476

QUESTION ID

S66-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. stress level

GRADING NOTE Stress can affect both exercise and sleep.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

True or false?

Random assignment controls lurking variables on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

2477

QUESTION ID

S66-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Balances groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

Causation can be concluded from

A. an observational study

B. a correlation alone

C. a randomized experiment

D. a survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

2478

QUESTION ID

S66-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B3-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does a correlation alone not show which variable causes the other?

A. Correlation can be found only for categories, not for numbers.

B. Correlation is always 0 when one variable causes the other.

C. Correlated variables can never be related to each other.

D. Either variable could affect the other, or a third variable could affect both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

2479

QUESTION ID

S66-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Either variable could affect the other, or a third variable could affect both.

GRADING NOTE Direction and lurking variables. The others are false: correlation is for numerical variables, is not 0 under causation, and correlated variables can be related.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 1

Improve an experimental design • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers try a new diet and lose weight. Why is a control group (for example a second, randomly assigned diet) needed to judge the diet?

A. Without a comparison group, the weight loss could be due to other factors, not the diet.

B. A control group makes every volunteer in the study lose more weight on the diet.

C. A control group proves that the new diet works for everyone who tries it.

D. A control group is needed only in studies with more than 100 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

2481

QUESTION ID
S66-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Without a comparison group, the weight loss could be due to other factors, not the diet.

GRADING NOTE Comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug study blocks subjects by age, then randomly assigns treatments within each age group. Why?

A. so the older subjects all get the drug and the younger the placebo

B. so every subject in each age group receives both treatments

C. so age differences cannot hide or create a treatment effect

D. so the study can avoid random assignment within the groups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

2482

QUESTION ID

S66-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. so age differences cannot hide or create a treatment effect

GRADING NOTE Blocking compares treatments within similar ages.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

True or false?

In a double-blind experiment, neither subjects nor evaluators know who gets the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

2483

QUESTION ID

S66-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets, and diet A led to more weight loss. Which change would let the result be generalized to all adults?

A. measuring the volunteers more often during the study

B. using a placebo for one group

C. adding more volunteers to each diet

D. randomly selecting subjects from all adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

2484

QUESTION ID

S66-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. randomly selecting subjects from all adults

GRADING NOTE Random sampling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

2484

Why are many subjects better than few in an experiment?

A. More subjects remove the need for random assignment to the groups.

B. More subjects always make the treatment itself work better.

C. More subjects make any observed result statistically significant.

D. Chance differences between groups are smaller, so real effects are easier to detect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

2485

QUESTION ID

S66-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Chance differences between groups are smaller, so real effects are easier to detect.

GRADING NOTE Less chance variation. The others are false: random assignment is still needed, size does not change the treatment, and not every result becomes significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 2

Use re-randomization to judge a treatment difference • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets; diet A's group lost 2.4 kg more on average. Under the model that the diets have the same effect, 5 of 200 re-randomizations gave a difference of at least 2.4 kg in favor of A. Use 5% as the cutoff for unusual. Is the difference significant?

A. No: 5/200 = 2.5% < 5%. **B.** Yes: 5/200 = 2.5% < 5%.

C. No: 5/200 = 25% > 5%. **D.** Yes: 5 is more than 2.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

2487

QUESTION ID
S66-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Yes:
5/200 = 2.5% < 5%.

GRADING NOTE 2.5% is below the cutoff.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

60 volunteers (not a random sample of adults) were randomly assigned to two diets, and diet A's group lost significantly more weight. Which conclusion is justified?

A. Diet A is associated with weight loss, but the study cannot show that it caused it.

B. There is evidence that diet A caused more weight loss for these volunteers; it may not apply to all adults.

C. Nothing can be concluded, because the volunteers were not a random sample of adults.

D. Diet A causes more weight loss for all adults, because the result was significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

2488

QUESTION ID
S66-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. There is evidence that diet A caused more weight loss for these volunteers; it may not apply to all adults.

GRADING NOTE Random assignment supports cause; no random sample limits the scope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

True or false?

Re-randomization shuffles the group labels as if the treatment had no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

2489

QUESTION ID
S66-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 70 of 200 re-randomized differences were at least 2.4 kg, conclude

A. the difference could be due to chance

B. the study was observational

C. diet B works

D. diet A works

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

2490

QUESTION ID
S66-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. the difference could be due to chance

GRADING NOTE 35% is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

2490

ANSWER KEY Reduced copy of the question page, with its answer

Why is a re-randomization test valid only when the original groups were formed by random assignment?

A. Without random assignment the group means cannot be computed, so there is nothing to shuffle in the test.

B. Random assignment makes every difference significant, so the test can only be used after random assignment.

C. Re-randomization needs exactly the same number of subjects as a fair coin needs flips, so it requires random groups.

D. The shuffle copies the random assignment; if the groups were not randomly formed, the shuffle does not model how they arose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

2491

QUESTION ID

S66-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The shuffle copies the random assignment; if the groups were not randomly formed, the shuffle does not model how they arose.

GRADING NOTE The simulation models the assignment process. The others are false: group means can always be computed, random assignment does not make results significant, and no coin is involved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population proportion with a margin of error from a simulation

MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Statistics, parameters, and sampling variability	**HSS-IC.A.1**
BACKWARD 2	Proportions and percents	**6.RP.A.3.c**
BACKWARD 3	Standard deviation as typical distance from the mean	**HSS-ID.A.2**
FORWARD 1	Plan a sample size for a desired margin of error	**HSS-IC.B.4**
FORWARD 2	Evaluate reports that use margins of error	**HSS-IC.B.6**

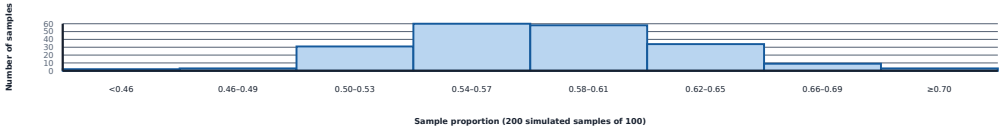
SET 67 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58 want a later start time. The histogram shows 200 simulated samples of 100 from a population with p = 0.58; their standard deviation is 0.049. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval estimate, as decimals.



Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

2494

QUESTION ID
S67-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.098; 0.482; 0.678

GRADING NOTE Form required: all three. 2(0.049) = 0.098; 0.58 ± 0.098.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 300 voters, 120 support a measure. Simulated sample proportions (n = 300, p = 0.40) have standard deviation 0.0275. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the estimate, the margin of error, and the interval, as decimals.

Estimate: _____

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

2495

QUESTION ID

S67-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.4; 0.055; 0.345; 0.455

GRADING NOTE Form required: all four. 120/300 = 0.40; 2(0.0275) = 0.055.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 shoppers, 35 use a store app. Simulated sample proportions (n = 50, p = 0.70) have standard deviation 0.064. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, as decimals.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

2496

QUESTION ID

S67-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.128; 0.572; 0.828

GRADING NOTE Form required: all three. 2(0.064) = 0.128; 0.70 ± 0.128.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Three random-sample surveys have simulated-proportion standard deviations 0.064 (n = 50), 0.049 (n = 100), and 0.0275 (n = 300). Using margin of error = 2 SD, which survey has the smallest margin of error, and why?

A. n = 100 (margin 0.098): the middle size balances the two kinds of error.

B. All are equal: the margin depends only on the population size.

C. n = 300 (margin 0.055): larger samples give less variable sample proportions.

D. n = 50 (margin 0.128): smaller samples are easier to measure exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

2497

QUESTION ID

S67-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. n = 300 (margin 0.055): larger samples give less variable sample proportions.

GRADING NOTE 2(0.0275) = 0.055 is smallest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

A report from a random sample says "52% of voters support the plan, with a margin of error of 3 percentage points (95% confidence)." Which interpretation is correct?

A. The percent of all voters who support the plan is plausibly between 49% and 55%.

B. Exactly 52% of all voters support the plan, give or take nothing at all.

C. The plan will win, because 52% is more than 50% of the voters.

D. Between 49% and 55% of the sampled voters support the plan.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

2498

QUESTION ID
S67-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. The percent of all voters who support the plan is plausibly between 49% and 55%.

GRADING NOTE The interval estimates the population percent; it includes values below 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 1

Statistics, parameters, and sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58 want a later start time. What is the statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

2500

QUESTION ID
S67-B1-Q1

TYPE
Short answer

CORRECT ANSWER
0.58 (58%)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

2500

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58% want a later start time. What parameter does 58% estimate?

A. the margin of error of the survey

B. the percent of all students who want a later start

C. the number of students in the sample

D. the percent of the 100 sampled students who want it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

2501

QUESTION ID

S67-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. the percent of all students who want a later start

GRADING NOTE The parameter describes the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

True or false?

Different random samples of 100 would give different sample proportions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

2502

QUESTION ID

S67-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram of 200 simulated sample proportions (samples of 100) shows

A. the population parameter exactly

B. how sample proportions vary

C. the whole population of values

D. one sample of 100 values

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

2503

QUESTION ID

S67-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. how sample proportions vary

GRADING NOTE Sampling distribution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

Why must a sample be random for its margin of error to be meaningful?

A. A random sample always has a margin of error of 0, so any error is bias.

B. A biased sample has a smaller margin of error than a random one.

C. The margin describes chance variation in random samples; a biased sample can be off by more.

D. The margin depends only on the size of the population, not the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

2504

QUESTION ID

S67-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The margin describes chance variation in random samples; a biased sample can be off by more.

GRADING NOTE The margin does not measure bias. The others are false: random samples have a positive margin, bias is not measured by the margin, and the margin depends on the sample size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

2504

SET 67 • HSS-IC.B.4

BACKWARD 2

Proportions and percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Write 120 out of 300 as a proportion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

2506

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.40

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.58 ± 0.10 as an interval.

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

2507

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.48; 0.68

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

35 out of 50 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

2508

QUESTION ID

S67-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35/50 = 0.70 = 70%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

52% ± 3% gives

A. 52% to 55%

B. 49% to 55%

C. 50% to 54%

D. 49% to 52%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

2509

QUESTION ID

S67-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 49% to 55%

GRADING NOTE Subtract and add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.055 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

2510

QUESTION ID

S67-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 3

Standard deviation as typical distance from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

By the 68-95-99.7 rule, what percent of a normal distribution lies within 2 standard deviations of the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

2512

QUESTION ID

S67-B3-Q1

TYPE

Short answer

CORRECT ANSWER

95%

GRADING NOTE The middle number of the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

If the SD is 0.049, what is 2 SDs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

2513

QUESTION ID

S67-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.098

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

True or false?

A smaller standard deviation means more variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

2514

QUESTION ID

S67-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A smaller SD means the values are closer to the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which set of simulated proportions varies least?

A. SD 0.1

B. SD 0.0275

C. SD 0.049

D. SD 0.064

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

2515

QUESTION ID

S67-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. SD 0.0275

GRADING NOTE Smallest SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B3-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 0.064.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

2516

QUESTION ID

S67-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.128

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 1

Plan a sample size for a desired margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A survey of a random sample of 100 students has a margin of error of 0.10. Quadrupling the sample size halves the margin of error. What is the margin of error with n = 400? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

2518

QUESTION ID

S67-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.05

GRADING NOTE 0.10/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

About what sample size gives a margin of error of 0.05 when $p \approx 0.5$? (Use margin $\approx 2\sqrt{(p(1-p)/n)}$.)

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

2519

QUESTION ID
S67-F1-Q2

TYPE
Short answer

CORRECT ANSWER
400

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Doubling the sample size halves the margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q3/5

2520

QUESTION ID
S67-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It divides it by √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of n = 100 people gives a margin of error for a proportion. Which change to the sample size reduces the margin of error the most?

A. use n = 900

B. use n = 200

C. use n = 110

D. take a second sample of 100 and report it alone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

2521

QUESTION ID
S67-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. use n = 900

GRADING NOTE The margin is about proportional to 1/√n: n = 900 divides it by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

Two random-sample polls of the same population have margins of error 0.05 and 0.025 for a proportion. The margin of error is about proportional to $1/\sqrt{n}$. How many times as large is the second sample as the first?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

2522

QUESTION ID

S67-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Halving the margin requires 4 times the sample size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 2

Evaluate reports that use margins of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A poll shows candidate A at 51% and B at 49%, each with a margin of error of ±4 percentage points. A headline says "A leads." Evaluate.

A. Justified: the margin of error is the same for both, so it cancels and A leads by 2.

B. Not justified: A must have less than 50% support, since 51% − 4% is below 50%.

C. Justified: 51% is greater than 49%, and the poll used a large sample.

D. Not justified: the intervals 47%−55% and 45%−53% overlap, so either could lead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

2524

QUESTION ID
S67-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. Not justified: the intervals 47%−55% and 45%−53% overlap, so either could lead.

GRADING NOTE The difference is within the margin of error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report gives 58% ± 10% from a random sample of 100 students and claims "a majority of students want a later start." Evaluate.

A. Not clearly justified: the interval 48% to 68% includes values below 50%.

B. Justified: 58% is more than 50%.

C. Justified: the whole interval is above 50%.

D. Not justified: a sample of 100 can never support a claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

2525

QUESTION ID
S67-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Not clearly justified: the interval 48% to 68% includes values below 50%.

GRADING NOTE Plausible values include less than a majority.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

True or false?

A margin of error accounts for bias from a voluntary-response sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

2526

QUESTION ID

S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which claim is best supported by its data?

A. A random sample gives 52% ± 4% support, so exactly 52% support it.

B. A random sample gives 70% ± 3% support, so a majority support the plan.

C. A random sample gives 51% ± 5% support, so a majority support the plan.

D. An online poll gives 60% support, so a majority support the plan.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2527

QUESTION ID
S67-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. A random sample gives 70% ± 3% support, so a majority support the plan.

GRADING NOTE Only 67%−73% stays above 50% and comes from a random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

Besides the percent, what should a careful poll report include?

A. the sample size, how the sample was chosen, and the margin of error

B. only the percent, rounded to the nearest whole number

C. only last year's result, so readers can compare

D. the names and answers of the people who were surveyed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

2528

QUESTION ID

S67-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. the sample size, how the sample was chosen, and the margin of error

GRADING NOTE Information needed to judge the estimate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population mean with a margin of error from a simulation

MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Mean of a data set	**6.SP.B.5.c**
BACKWARD 2	What resampling simulates	**HSS-IC.A.1**
BACKWARD 3	Two standard deviations and the 95% range	**HSS-ID.A.2**
FORWARD 1	Use intervals to judge claims about a mean	**HSS-IC.B.4**
FORWARD 2	Compare two group means with re-randomization	**HSS-IC.B.5**

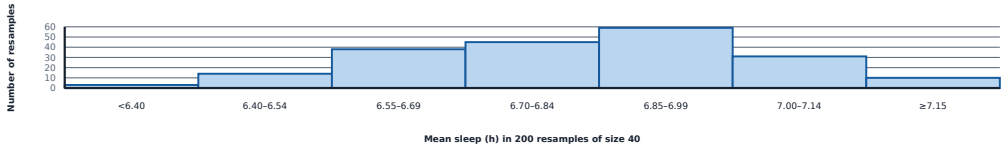
SET 68 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 40 students slept a mean of 6.8 hours. The histogram shows the means of 200 resamples of the data; their standard deviation is 0.20 hours. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in hours.



Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

2531

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.4; 6.4; 7.2

GRADING NOTE Form required: all three. 2(0.20) = 0.4; 6.8 ± 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 25 apples has mean weight 152 g. Simulated sample means have standard deviation 3.6 g. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in grams.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

2532

QUESTION ID
S68-M-Q2

TYPE
Short answer

CORRECT ANSWER
7.2; 144.8; 159.2

GRADING NOTE Form required: all three. 2(3.6) = 7.2; 152 ± 7.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 60 commuters has mean commute 24.5 minutes. Simulated sample means have standard deviation 1.4 minutes. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range). Give the margin of error and the interval, in minutes.

Margin of error: _____

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

2533

QUESTION ID

S68-M-Q3

TYPE

Short answer

CORRECT ANSWER

2.8; 21.7; 27.3

GRADING NOTE Form required: all three. 2(1.4) = 2.8; 24.5 ± 2.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

A random sample of 40 students gives the 95% interval 6.4 to 7.2 hours for the mean hours of sleep. Which interpretation is correct?

A. Every student at the school sleeps at least 6.4 hours each night.

B. The 40 sampled students each slept between 6.4 and 7.2 hours.

C. The mean sleep of all students is plausibly between 6.4 and 7.2 hours.

D. 95% of all students sleep between 6.4 and 7.2 hours each night.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

2534

QUESTION ID

S68-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The mean sleep of all students is plausibly between 6.4 and 7.2 hours.

GRADING NOTE The interval is about the population mean, not individual students.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 25 apples gives a margin of error of 7.2 g for the mean weight. Assume apple weights vary the same way and the same 2-SD (95%) convention is used, so the margin of error is proportional to 1/√n. What margin of error do you expect from a random sample of 100 apples? Give grams.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

2535

QUESTION ID
S68-M-Q5

TYPE
Short answer

CORRECT ANSWER
3.6

GRADING NOTE 4 times the sample size halves the margin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 1

Mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 6, 7, 7.5, 6.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

2537

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6.75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

What does the mean of a random sample estimate?

A. the margin of error

B. the population mean

C. the largest value in the population

D. the sample size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

2538

QUESTION ID

S68-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. the population mean

GRADING NOTE Statistic estimates parameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The mean is not affected by extreme values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

2539

QUESTION ID

S68-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One extreme value can move the mean a lot; the median is resistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

2539

ANSWER KEY Reduced copy of the question page, with its answer

The mean of 150, 152, 154 is

A. 154

B. 150

C. 152

D. 456

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

2540

QUESTION ID

S68-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 152

GRADING NOTE Sum/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

2540

ANSWER KEY Reduced copy of the question page, with its answer

Each of 40 values in a data set increases by 0.5. By how much does the mean increase? Give a number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

2541

QUESTION ID

S68-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE Adding a constant to every value adds it to the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 2

What resampling simulates • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why can resampling a random sample (drawing with replacement) show how sample means vary?

A. Resampling collects new data from the whole population each time it is done.

B. Resampling always gives the true population mean as the center.

C. Resampling removes all chance variation from the sample means.

D. The resamples act like new samples from a population that looks like the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

2543

QUESTION ID

S68-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. The resamples act like new samples from a population that looks like the sample.

GRADING NOTE The sample stands in for the population. The others are false: resampling uses only the original sample, does not give the true mean, and does not remove chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a histogram of 200 resample means, what does one bar's height show?

A. the chance the population mean is in that interval

B. the mean of the original sample

C. how many of the 200 resample means fell in that interval

D. the number of values in each resample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

2544

QUESTION ID

S68-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. how many of the 200 resample means fell in that interval

GRADING NOTE Bar height = count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Sample means vary less than individual values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

2545

QUESTION ID

S68-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Averaging reduces variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 40 students slept a mean of 6.8 hours. The parameter being estimated is

A. 6.8

B. the mean sleep of all students

C. the 40 students' mean

D. the resample SD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

2546

QUESTION ID

S68-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the mean sleep of all students

GRADING NOTE Population mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

A random sample of 40 students had a mean of 6.8 hours of sleep. Is 6.8 hours a parameter or a statistic?

A. a statistic: it describes the sample of 40

B. a parameter: it describes all students

C. a statistic: it describes all students

D. a parameter: it describes the sample of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

2547

QUESTION ID

S68-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. a statistic: it describes the sample of 40

GRADING NOTE It is computed from the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 3

Two standard deviations and the 95% range • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute 6.8 − 2(0.20) and 6.8 + 2(0.20).

6.8 − 2(0.20) = _____

6.8 + 2(0.20) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

2549

QUESTION ID

S68-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6.4; 7.2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 1.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

2550

QUESTION ID

S68-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

True or false?

About 95% of a mound-shaped distribution is within 2 SDs of its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

2551

QUESTION ID

S68-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

152 ± 7.2 gives

A. 144.8 to 159.2

B. 145 to 159

C. 152 to 159.2

D. 144.8 to 152

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

2552

QUESTION ID

S68-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 144.8 to 159.2

GRADING NOTE Subtract and add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 7.5 inside the interval from 6.4 to 7.2? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

2553

QUESTION ID

S68-B3-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE 7.5 > 7.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 1

Use intervals to judge claims about a mean • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A random sample gives the 95% interval 6.4 to 7.2 hours for students' mean sleep. A health guide says students average 8 hours. Does the interval support this?

A. Yes: some of the students in the sample sleep 8 hours or more.

B. No: 8 is outside the interval of plausible values for the mean.

C. Yes: 8 is close to the upper end of the interval, 7.2 hours.

D. No: the interval proves the mean is exactly its center, 6.8 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

2555

QUESTION ID

S68-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. No: 8 is outside the interval of plausible values for the mean.

GRADING NOTE 8 is not a plausible mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

A random sample gives the 95% interval 144.8 g to 159.2 g for the mean apple weight. A grower claims the apples average 155 g. Is the claim consistent with the interval?

A. No: the mean must be exactly 152 g.

B. Yes: the claim is proved true because 155 is inside it.

C. No: 155 g is not the center of the interval.

D. Yes: 155 g is inside the interval, so it is a plausible mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

2556

QUESTION ID

S68-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Yes: 155 g is inside the interval, so it is a plausible mean.

GRADING NOTE Consistent, not proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A value inside a 95% interval estimate for a mean is proved to be the true mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

2557

QUESTION ID

S68-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is only plausible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

A random sample of 60 commuters gives an interval for the mean commute time. Which change would most likely narrow the interval?

A. sampling 240 commuters instead of 60

B. using the median

C. rounding the data

D. sampling 30 commuters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

2558

QUESTION ID

S68-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. sampling 240 commuters instead of 60

GRADING NOTE Larger n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of students at one school gives the interval 6.4 to 7.2 hours for their mean sleep. Why should the interval not be applied to students at a different school?

A. The students at the other school must have a higher mean sleep time.

B. An interval for a mean can only describe a single student's sleep.

C. The interval is too narrow to be applied to any group of students.

D. The sample came only from this school, so it estimates only this school's mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

2559

QUESTION ID
S68-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. The sample came only from this school, so it estimates only this school's mean.

GRADING NOTE The population sampled limits the inference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 2

Compare two group means with re-randomization • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, 40 students were randomly assigned to use a study app or not (20 each). The app group scored 6 points more on average. Under the model that the app has no effect, 9 of 200 re-randomizations gave the app group at least 6 more points. Using a one-sided count and a 5% cutoff, is the difference significant?

A. No: 9/200 = 4.5% > 5%.

B. Yes: 9/200 = 4.5% < 5%.

C. No: 9 is more than 6.

D. Yes: 9/200 = 45%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

2561

QUESTION ID

S68-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Yes:

9/200 = 4.5% < 5%.

GRADING NOTE 4.5% is below the cutoff.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

Students were randomly assigned to a treatment group (study app) or a control group (no app), and the difference in mean scores was re-randomized 200 times. What null model do the re-randomizations represent?

A. The app has no effect on scores, so the group labels are interchangeable.

B. The app raises every score by exactly 6 points.

C. The students were chosen by voluntary response.

D. The control group scores lower in every re-randomization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

2562

QUESTION ID
S68-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. The app has no effect on scores, so the group labels are interchangeable.

GRADING NOTE Null model: no treatment effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

An interval for one mean and a re-randomization test for two means both use simulated variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

2563

QUESTION ID
S68-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Same idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

In an app experiment, the app group scored 6 points more on average. Under the model of no app effect, 30 of 200 re-randomizations gave the app group at least 6 more points. Use 5% as the cutoff for unusual. The conclusion is

A. the difference could be chance (15% > 5%)

B. the app works (15% < 50%)

C. the app is proved to have no effect

D. the app lowers scores

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

2564

QUESTION ID

S68-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the difference could be chance (15% > 5%)

GRADING NOTE 30/200 = 15%. Not significant is not proof of no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

2564

ANSWER KEY Reduced copy of the question page, with its answer

How is the margin of error for one mean related to judging a difference between two means?

A. A difference between two means never varies by chance, so it does not need any measure of chance variation.

B. The margin of error for one mean is always equal to the difference between the two group means.

C. Both ask how much a statistic varies by chance; a difference is meaningful only if it is large compared with that variability.

D. They are unrelated: one uses the data, and the other uses only the number of subjects in each group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

2565

QUESTION ID
S68-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. Both ask how much a statistic varies by chance; a difference is meaningful only if it is large compared with that variability.

GRADING NOTE Both compare a statistic with its chance variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

SET 69 • HSS-IC.B.5 • STATISTICS, INFERENCE, AND DECISIONS

Compare two treatments with a re-randomization simulation

MAIN	Algebra II main question	**HSS-IC.B.5**
BACKWARD 1	Compare the means of two groups	**7.SP.B.4**
BACKWARD 2	Random assignment in experiments	**HSS-IC.B.3**
BACKWARD 3	Probability as relative frequency in simulations	**7.SP.C.6**
FORWARD 1	Two-sided comparisons and design choices	**HSS-IC.B.5**
FORWARD 2	Evaluate reports of experiments	**HSS-IC.B.6**

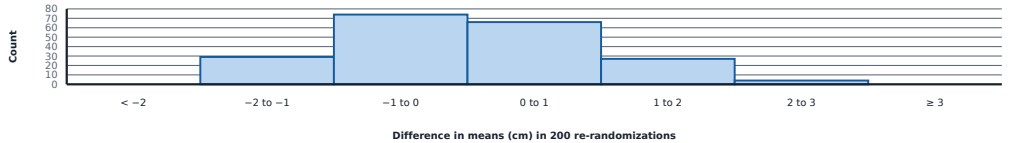
SET 69 • HSS-IC.B.5

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

10 plants got fertilizer and 10 did not (randomly assigned). Fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences under the model that fertilizer has no effect. Use a one-sided count (2.9 cm or more) and a 5% cutoff. Which conclusion is justified?



A. Most re-randomized differences were near 0, so the observed difference is not significant.

B. 4 of 200 (2%) were 2 cm or more, so the observed difference of 2.9 cm is not significant.

C. 0 of 200 were 2.9 cm or more; 0% < 5%, so the difference is significant: fertilizer appears to increase growth.

D. 0 of 200 were 2.9 cm or more, so this difference has probability exactly 0 and fertilizer is proved to work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

2568

QUESTION ID

S69-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 0 of 200 were 2.9 cm or more; 0% < 5%, so the difference is significant: fertilizer appears to increase growth.

GRADING NOTE The ≥ 3 bar and the part of the 2-to-3 bar at 2.9 or more: none of 200. Rare is not impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a randomized caffeine experiment, the caffeine group's mean reaction time was 0.04 s faster. Under the model that caffeine has no effect, 62 of 200 re-randomizations gave the caffeine group a mean at least 0.04 s faster. Using a one-sided count and a 5% cutoff, is the difference significant?

A. No: 62/200 = 31% > 5%, so the difference could easily be chance.

B. Yes: 62/200 = 31% is more than 5%, so the difference is significant.

C. No: 31% > 5%, and this proves that caffeine has no effect.

D. Yes: 0.04 s is less than 5% of a second, so it is significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

2569

QUESTION ID
S69-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. No: 62/200 = 31% > 5%, so the difference could easily be chance.

GRADING NOTE 31% is common. Not significant is not proof of no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

Students were randomly assigned to use a study app or not, and the app group scored 6 points higher on average. Under the model that the app has no effect, 9 of 200 re-randomizations gave the app group at least 6 more points. Using a one-sided count and a 5% cutoff, is the difference significant?

A. Yes: 9/200 = 45%, which is above 5%.

B. No: 9 is more than 6, so the app has no effect.

C. No: 9/200 = 4.5%, which is above 5%.

D. Yes: 9/200 = 4.5%, which is below 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

2570

QUESTION ID

S69-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Yes: 9/200 = 4.5%, which is below 5%.

GRADING NOTE One-sided: 9/200 = 4.5% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What does a re-randomization simulation assume, and how is it carried out?

A. It assumes no treatment effect; each group's values are re-sorted from smallest to largest, and the two sorted lists are compared once to find the difference.

B. It assumes the treatment works; the observed difference is doubled and halved many times, and the results are graphed to show how large the effect could be.

C. It assumes no treatment effect; all responses are pooled and randomly re-split into groups of the original sizes, and the difference in means is recorded many times.

D. It assumes the treatment works; new subjects are recruited for each run, and the difference in their group means is recorded many times to estimate the true effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

2571

QUESTION ID

S69-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. It assumes no treatment effect; all responses are pooled and randomly re-split into groups of the original sizes, and the difference in means is recorded many times.

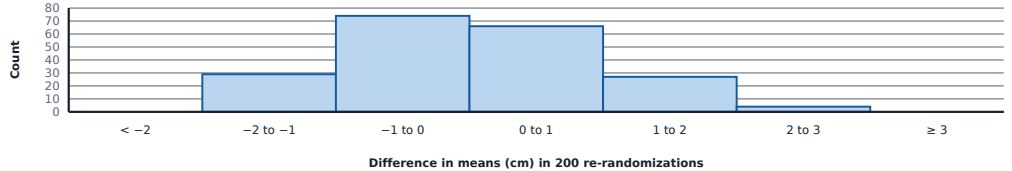
GRADING NOTE Null model plus shuffling of labels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the histogram of 200 re-randomized differences, how many differences were 2 cm or more, and would an observed difference of 2 cm be significant using a one-sided count and a 5% cutoff?



A. 0 of 200; a difference of 2 cm has probability 0 under the model.

B. 4 of 200 (2%); 2% < 5%, so yes, it would be significant.

C. 31 of 200 (15.5%); 15.5% > 5%, so no, it would not be significant.

D. 4 of 200 (2%); 2% < 5%, so no, it would not be significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

2572

QUESTION ID

S69-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 4 of 200 (2%); 2% < 5%, so yes, it would be significant.

GRADING NOTE The 2-to-3 bar has 4 and the ≥ 3 bar has 0. 31 counts from 1 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 1

Compare the means of two groups • Grade 7 • Statistics & Probability • 7.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Group A has mean 25.4 cm and group B has mean 22.5 cm. Find the difference A − B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

2574

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.9 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Group A: 4, 6, 8. Group B: 3, 5, 4. Find the difference in means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

2575

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

True or false?

A difference in sample means could happen even if the treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

2576

QUESTION ID

S69-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Mean A = 0.52 s and mean B = 0.48 s. A − B is

A. −0.04 s

B. 0.4 s

C. 1.00 s

D. 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

2577

QUESTION ID

S69-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.04 s

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is a difference between two group means compared with the variability in the groups?

A. Any difference between the group means proves a treatment effect.

B. The variability in the groups gives the exact size of the effect.

C. A difference is meaningful only if it is large relative to the natural variation.

D. Variability matters only when the two group means are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

2578

QUESTION ID

S69-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. A difference is meaningful only if it is large relative to the natural variation.

GRADING NOTE Signal versus noise. The others are false: a difference alone is not proof, variability is not the effect size, and variability always matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 2

Random assignment in experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

In a fertilizer experiment, 20 plants were randomly assigned to fertilizer or no fertilizer. Why were they randomly assigned?

A. to make the groups comparable, so a difference can be due to the fertilizer

B. so that no measurements of growth are needed afterward

C. so the result can be generalized to all plants of every kind, everywhere

D. to make sure every plant grows exactly the same amount

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

2580

QUESTION ID

S69-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. to make the groups comparable, so a difference can be due to the fertilizer

GRADING NOTE Random assignment balances other factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How can 20 plants be randomly assigned to two groups of 10?

A. Put the 10 tallest plants in the fertilizer group.

B. Let the gardener pick 10 healthy-looking plants for fertilizer.

C. Put the 10 plants nearest the window in the fertilizer group.

D. Number the plants 1-20 and use random numbers to choose 10 for fertilizer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

2581

QUESTION ID

S69-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Number the plants 1-20 and use random numbers to choose 10 for fertilizer.

GRADING NOTE Chance alone decides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Without random assignment, a re-randomization test does not justify a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

2582

QUESTION ID

S69-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Design matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Random assignment is a feature of

A. censuses

B. experiments

C. observational studies

D. surveys

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

2583

QUESTION ID

S69-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. experiments

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A randomized experiment gives a significant difference. Which conclusion does it allow?

A. the difference is due to chance alone, not the treatment

B. there is evidence the treatment caused the difference for these subjects

C. the treatment causes the same difference in everyone everywhere

D. the subjects were a random sample of everyone in the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

2584

QUESTION ID

S69-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. there is evidence the treatment caused the difference for these subjects

GRADING NOTE Cause for the subjects studied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 3

Probability as relative frequency in simulations • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

62 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

2586

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

31%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

9 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

2587

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

True or false?

An event that happens in 40% of simulated runs is unusual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

2588

QUESTION ID

S69-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 40% is common; unusual usually means about 5% or less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

0 of 200 runs suggests the observed result is

A. very unlikely under the model

B. certain

C. impossible

D. typical

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

2589

QUESTION ID

S69-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. very unlikely under the model

GRADING NOTE Rare, not impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 4/200 less than 5%? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

2590

QUESTION ID

S69-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE 4/200 = 2%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

2590

SET 69 • HSS-IC.B.5

FORWARD 1

Two-sided comparisons and design choices • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In a caffeine experiment, the observed difference in mean reaction time was 0.04 s. The question is whether caffeine changes reaction time in either direction. Which re-randomized differences should be counted?

A. only those with difference ≥ 0.04 s

B. those with |difference| ≥ 0.04 s, in either direction

C. only those with |difference| ≤ 0.04 s

D. only those with difference exactly 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

2592

QUESTION ID

S69-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. those with |difference| ≥ 0.04 s, in either direction

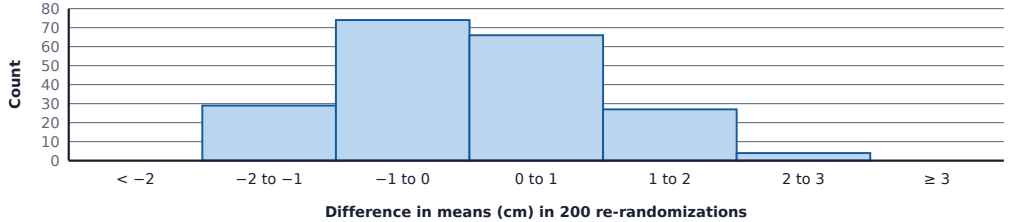
GRADING NOTE A two-sided question counts both tails.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows 200 re-randomized differences in mean growth (cm). How many differences were 2 cm or more, or less than −2 cm?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

2593

QUESTION ID
S69-F1-Q2

TYPE
Short answer

CORRECT ANSWER
4

GRADING NOTE The 2-to-3 bar has 4; the ≥ 3 and < −2 bars have 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

Using more plants in each group usually makes the re-randomized differences less spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

2594

QUESTION ID
S69-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Less chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

A small observed difference can be significant when

A. the groups have very little natural variation

B. the data are rounded to whole numbers

C. the groups were not randomly assigned

D. the sample in each group is small

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

2595

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the groups have very little natural variation

GRADING NOTE Small spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why should the cutoff for significance (such as 5%) be chosen before seeing the data?

A. The cutoff changes the data that are collected in the study.

B. The cutoff must equal the observed proportion of the data.

C. Choosing it afterward could bias the conclusion toward the result one wants.

D. Cutoffs chosen in advance are always 50%, the fairest choice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

2596

QUESTION ID

S69-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Choosing it afterward could bias the conclusion toward the result one wants.

GRADING NOTE Avoids bias. The others are false: the cutoff does not change the data, is not set equal to the result, and is usually 5%, not 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

2596

SET 69 • HSS-IC.B.5

FORWARD 2

Evaluate reports of experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A company says its supplement "improved memory scores by 3 points" in a study of 8 people with no control group. Which evaluation is correct?

A. Strong evidence: an increase of 3 points is a real improvement in memory for these people.

B. Weak evidence, but only because an increase of 3 points is too small a number to matter.

C. Strong evidence: every one of the 8 people in the study took the supplement as directed.

D. Weak evidence: with no control group and only 8 people, the change could be due to other factors or chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

2598

QUESTION ID
S69-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. Weak evidence: with no control group and only 8 people, the change could be due to other factors or chance.

GRADING NOTE No comparison and a tiny sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says a new teaching method "significantly" raised scores in a randomized experiment. Under the no-effect model, 2 of 200 re-randomized differences were at least as large as the observed difference. Use 5% as the cutoff for unusual. Evaluate the report.

A. Supported: the assignment was random, and 2/200 = 1% < 5%.

B. Supported: the method is proved to work for all students.

C. Not supported: re-randomization cannot be used for experiments.

D. Not supported: 2/200 = 1% is too small to matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

2599

QUESTION ID

S69-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Supported: the assignment was random, and 2/200 = 1% < 5%.

GRADING NOTE Significant evidence from a randomized design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"Statistically significant" means the effect is large and important.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

2600

QUESTION ID
S69-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It means unlikely by chance; it may be small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

Which report is most convincing?

A. a study of 500 users with no comparison group or random assignment

B. testimonials from many satisfied customers

C. a randomized experiment with a control group and a rare result

D. an online poll of 10,000 users of the product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

2601

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment with a control group and a rare result

GRADING NOTE Design and evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

Which information should a report of an experiment include?

A. only the observed difference between the two groups

B. the assignment method, group sizes, observed difference, and how unusual it is

C. only the conclusion that the treatment works for everyone

D. only the names of the treatments, the date of the study, and who paid for it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

2602

QUESTION ID
S69-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. the assignment method, group sizes, observed difference, and how unusual it is

GRADING NOTE All the evidence needed to judge it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSS-IC.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Evaluate claims in reports based on data

MAIN	Algebra II main question	**HSS-IC.B.6**
BACKWARD 1	Random samples and bias	**HSS-IC.A.1**
BACKWARD 2	Study design and causation	**HSS-IC.B.3**
BACKWARD 3	Read scales on graphs	**3.MD.B.3**
FORWARD 1	Rewrite claims so that they follow from the data	**HSS-IC.B.6**
FORWARD 2	Analyze decisions based on test results	**HSS-MD.B.7**

SET 70 • HSS-IC.B.6

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An ad says "4 out of 5 dentists recommend our toothpaste." Only 5 dentists, chosen by the company, were asked. Evaluate the claim.

A. Weak: the sample is tiny and not random, so it cannot represent all dentists.

B. Strong: dentists are experts, so any group of them is representative.

C. Weak: the claim would only be fine if all 5 of the dentists agreed.

D. Strong: 80% is a large majority of the dentists who were asked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

2605

QUESTION ID

S70-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Weak: the sample is tiny and not random, so it cannot represent all dentists.

GRADING NOTE Small, company-chosen sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "Students who eat breakfast earn higher grades, so breakfast causes higher grades." The data came from a survey. Evaluate.

A. Justified: the survey found that students who eat breakfast have higher grades.

B. Not justified: breakfast and grades cannot be associated in a survey.

C. Not justified: a survey shows association, not causation; lurking variables could explain it.

D. Justified: the survey was large enough to show cause and effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

2606

QUESTION ID
S70-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Not justified: a survey shows association, not causation; lurking variables could explain it.

GRADING NOTE Observational data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph shows sales of 50 and 52 thousand dollars, with the vertical axis starting at 48 thousand, so one bar looks twice as tall. Evaluate the impression the graph gives.

A. Fair: starting the axis at 48 thousand makes the graph easier to read.

B. Fair: the tops of the bars show the correct values of 50 and 52 thousand.

C. Misleading: the taller bar should be drawn more than twice as tall.

D. Misleading: the truncated axis makes a 4% difference look like a 100% difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

2607

QUESTION ID

S70-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Misleading: the truncated axis makes a 4% difference look like a 100% difference.

GRADING NOTE 52 is 4% more than 50, but the bars above 48 are 2 and 4 units tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A TV show asks viewers to text their vote; 90% of 10,000 texts favor a policy. The show claims most citizens support it. Evaluate.

A. Justified: 90% is so far above 50% that the method of collecting the votes does not matter.

B. Not justified: texters are a voluntary response, not a random sample.

C. Justified: 10,000 texts form a very large sample of the citizens.

D. Not justified: 10,000 texts is too few to represent a whole country.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

2608

QUESTION ID
S70-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Not justified: texters are a voluntary response, not a random sample.

GRADING NOTE Voluntary response bias is not fixed by size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

A poll shows candidate A at 51% and B at 49%, each with a margin of error of ±4 percentage points. A headline says "A is winning." Evaluate.

A. Not justified: the intervals 47%−55% and 45%−53% overlap, so the lead is within the margin of error.

B. Not justified: B is actually winning, because the margin favors the trailing candidate.

C. Justified: the margin of error is the same for both candidates, so it cancels out.

D. Justified: A has more support than B in the poll, so A is winning the race.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

2609

GRADING NOTE Either could lead.

QUESTION ID

S70-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Not justified: the intervals 47%−55% and 45%−53% overlap, so the lead is within the margin of error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 1

Random samples and bias • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why is a voluntary-response sample usually biased?

A. Every member of the population responds to it.

B. Voluntary samples are always too small to use.

C. The sample is chosen with random numbers instead.

D. People with strong opinions are more likely to respond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

2611

QUESTION ID

S70-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. People with strong opinions are more likely to respond.

GRADING NOTE Self-selection. The others are false: not everyone responds, size is not the problem, and volunteers are not chosen with random numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What does a random sample allow?

A. concluding cause and effect from the data

B. skipping the margin of error in the report

C. generalizing from the sample to the population

D. eliminating all chance variation from results

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

2612

QUESTION ID

S70-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. generalizing from the sample to the population

GRADING NOTE Generalization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A very large sample cannot be biased.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

2613

QUESTION ID

S70-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not fix bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

Which sample is most likely biased?

A. customers who chose to write reviews

B. a random sample of 1,000

C. randomly chosen customers

D. every customer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

2614

QUESTION ID

S70-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. customers who chose to write reviews

GRADING NOTE Self-selected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is "5 dentists chosen by the company" a random sample? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

2615

QUESTION ID

S70-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 2

Study design and causation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A survey records students' breakfast habits and grades. What type of study is it?

A. an observational study

B. a randomized experiment

C. a placebo-controlled experiment

D. a simulation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

2617

QUESTION ID

S70-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. an observational study

GRADING NOTE No treatment is assigned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which design could test whether eating breakfast causes higher grades?

A. a voluntary online poll about eating breakfast and grades

B. a larger survey of students' breakfast habits and grades

C. a randomized experiment that assigns students to eat breakfast or not

D. comparing grades of students who already eat breakfast or not

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

2618

QUESTION ID

S70-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment that assigns students to eat breakfast or not

GRADING NOTE Random assignment is needed for causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

True or false?

Only randomized experiments can establish cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

2619

QUESTION ID

S70-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A confounding variable is

A. the response variable that is measured in the study

B. a variable related to both the explanatory and response variables

C. the number of subjects in the sample of the study

D. the margin of error of the estimate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

2620

QUESTION ID
S70-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. a variable related to both the explanatory and response variables

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

2620

Students who eat breakfast tend to earn higher grades. Which is a plausible confounding variable?

A. the students' locker numbers

B. the amount of sleep students get

C. the color of the cereal box

D. the day the survey was printed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

2621

QUESTION ID

S70-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. the amount of sleep students get

GRADING NOTE Sleep can affect both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 3

Read scales on graphs • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph's vertical axis starts at 48. One bar reaches 50 and another reaches 52. What values do the two bars show?

First bar: _____

Second bar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

2623

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

50; 52

GRADING NOTE Form required: both. Read the scale, not the bar length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent larger is 52 than 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

2624

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

True or false?

An axis that does not start at 0 can make differences look larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

2625

QUESTION ID

S70-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Truncation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

On a graph whose axis starts at 48, a bar of 52 looks how many times as tall as a bar of 50?

A. the same

B. 1.04 times

C. 2 times

D. 4 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

2626

QUESTION ID

S70-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2 times

GRADING NOTE Heights 4 and 2 above the axis start.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

A bar graph of sales of 50 and 52 thousand dollars has a vertical axis starting at 48 thousand. How should it be redrawn to be fair?

A. Make the bars wider.

B. Remove the axis labels.

C. Start the axis at 51 to show the difference better.

D. Start the vertical axis at 0, or clearly mark the axis break.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

2627

QUESTION ID

S70-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Start the vertical axis at 0, or clearly mark the axis break.

GRADING NOTE A truncated axis exaggerates differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 1

Rewrite claims so that they follow from the data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A survey found that students who eat breakfast earn higher grades. Which claim is supported by the survey?

A. Higher grades cause students to eat breakfast.

B. Eating breakfast causes higher grades.

C. Students who eat breakfast tend to have higher grades.

D. Skipping breakfast causes lower grades.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

2629

QUESTION ID

S70-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Students who eat breakfast tend to have higher grades.

GRADING NOTE An observational survey supports only an association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A poll shows 51% for A and 49% for B, each with a margin of error of ±4 percentage points. Which headline is accurate?

A. "The race is too close to call"

B. "B is winning"

C. "A is winning"

D. "A has a certain 2-point lead"

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

2630

QUESTION ID

S70-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. "The race is too close to call"

GRADING NOTE The difference is within the margin of error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

2630

True or false?

Saying "is associated with" instead of "causes" is appropriate for an observational study.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

2631

QUESTION ID

S70-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Careful wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which conclusion is justified by a random sample of 1,000 voters showing 62% ± 3% support?

A. the sample was biased

B. most voters support the measure

C. exactly 62% support it

D. the measure will pass by 24 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

2632

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. most voters support the measure

GRADING NOTE 59% to 65%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

An ad says "4 out of 5 dentists recommend our toothpaste," based on 5 dentists chosen by the company. What would make the claim more convincing?

A. printing the claim in larger letters on the box

B. a large random sample of dentists, with the question wording reported

C. asking the same 5 dentists again a year later

D. choosing 5 different dentists who already like the brand

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

2633

QUESTION ID

S70-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. a large random sample of dentists, with the question wording reported

GRADING NOTE Size and random selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 2

Analyze decisions based on test results • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A screening test detects 95% of people who have a disease (sensitivity 95%) and gives a false positive for 5% of healthy people. In a group of 10,000 people, 1% have the disease. How many true positives are there, and how many positive results in all?

True positives: _____

All positives: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

2635

QUESTION ID

S70-F2-Q1

TYPE

Short answer

CORRECT ANSWER

95; 590

GRADING NOTE Form required: both. 0.95(100) = 95; 0.05(9,900) = 495; 95 + 495 = 590.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A screening test has sensitivity 95% and a 5% false-positive rate. In a group of 10,000 people where 1% have the disease, there are 95 true positives and 495 false positives. A report says "a positive test means you almost surely have the disease." Evaluate.

A. Correct: only 5% of positive results are false.

B. Misleading: 495/590 ≈ 84% of positives have the disease.

C. Correct: the test detects 95% of cases.

D. Misleading: only 95/590 ≈ 16% of positives have the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

2636

QUESTION ID

S70-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Misleading: only 95/590 ≈ 16% of positives have the disease.

GRADING NOTE P(disease | positive) = 95/590 ≈ 0.161.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

True or false?

The probability of having the disease given a positive result can be low even with an accurate test.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

2637

QUESTION ID
S70-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Rare disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A screening test has sensitivity 95% and a 5% false-positive rate. If 1% of a population has the disease, P(disease | positive) is about

A. 0.99

B. 0.05

C. 0.95

D. 0.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

2638

QUESTION ID
S70-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 0.16

GRADING NOTE Per 10,000: 95 true and 495 false positives; 95/590 ≈ 0.161.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For a screening test, P(disease | positive) ≈ 0.16. Which follow-up decision makes the most sense after a positive result?

A. Begin treatment at once, since the test detects 95% of cases.

B. Ignore the result, since it is probably a false positive.

C. Repeat the same test and treat only if it is negative.

D. Confirm with a second, more accurate test before treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

2639

QUESTION ID
S70-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. Confirm with a second, more accurate test before treatment.

GRADING NOTE Most positives are false, but some are real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSS-MD.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Design and judge fair decision procedures with probabilities

MAIN	Algebra II main question	**HSS-MD.B.6**
BACKWARD 1	Probabilities of equally likely outcomes	**7.SP.C.7.a**
BACKWARD 2	Sample spaces for compound events	**7.SP.C.8.b**
BACKWARD 3	Probability between 0 and 1 and redoing outcomes	**7.SP.C.5**
FORWARD 1	Design fair procedures and count expected trials	**HSS-MD.B.6**
FORWARD 2	Analyze strategies and decisions with probability	**HSS-MD.B.7**

SET 71 • HSS-MD.B.6

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Three friends choose a driver by flipping a fair coin twice: HH = Ana, HT = Ben, TH = Cy, and TT = flip twice again. Is this fair?

A. No: TT is ignored, so the three probabilities add to 3/4 instead of 1.

B. Yes: each friend has probability 1/4, the same as each of the four outcomes.

C. Yes: HH, HT, and TH are equally likely and TT is redone, so each friend has probability 1/3.

D. No: Ana has probability 1/2, because HH has two heads and the others have only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

2642

QUESTION ID
S71-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Yes: HH, HT, and TH are equally likely and TT is redone, so each friend has probability 1/3.

GRADING NOTE Redoing TT scales 1/4 each to 1/3 each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

Four people use a fair die: 1 = A, 2 = B, 3 = C, 4 = D, and 5 or 6 means roll again. Is this fair?

A. No: D is less likely, because 4 is next to 5 and 6.

B. Yes: each person has probability 1/4 after rerolls.

C. Yes: each person has probability 1/6.

D. No: each person has 1/6, and 1/3 of the time is wasted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

2643

QUESTION ID

S71-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Yes: each person has probability 1/4 after rerolls.

GRADING NOTE Four equally likely faces remain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A well-mixed hat holds 3 slips with Ana's name and 2 with Ben's. One slip is drawn at random to choose who goes first. Is this fair?

A. No: P(Ana) = 3/5 and P(Ben) = 2/5.

B. Yes: the slips are well mixed.

C. Yes: each name is drawn or not, so each has 1/2.

D. No: P(Ana) = 2/5 and P(Ben) = 3/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

2644

QUESTION ID
S71-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. No: P(Ana) = 3/5 and P(Ben) = 2/5.

GRADING NOTE Each slip is equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

A random number generator gives a whole number 1-100, each equally likely. Numbers 1-33 choose A, 34-66 choose B, 67-99 choose C, and 100 means generate again. Is this fair?

A. Yes: each has probability 33/100, and the leftover 1/100 does not matter.

B. Yes: each person has 33 equally likely numbers, and 100 is redone, so each has 1/3.

C. No: A has 33 numbers but C has 34 numbers, so C is a little more likely.

D. No: 100 is wasted, so the probabilities are 0.33 each and do not add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

2645

QUESTION ID
S71-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Yes: each person has 33 equally likely numbers, and 100 is redone, so each has 1/3.

GRADING NOTE Each range has 33 numbers; redoing 100 scales each to 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled; if the sum is even, player 1 wins, otherwise player 2 wins. Is this fair?

A. Yes: 18 of the 36 equally likely outcomes have an even sum, so each player has 1/2.

B. Yes: there are 11 possible sums, and about half of the 11 sums are even.

C. No: there are 6 even sums (2, 4, ..., 12) and only 5 odd sums, so player 1 wins more.

D. No: a sum of 7 is the most likely sum, so player 2 wins more often than player 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

2646

QUESTION ID
S71-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Yes: 18 of the 36 equally likely outcomes have an even sum, so each player has 1/2.

GRADING NOTE Count equally likely outcomes, not sums.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 1

Probabilities of equally likely outcomes • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. What is P(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

2648

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 3 red and 2 blue slips. What is P(red)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

2649

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

True or false?

Each face of a fair die has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

2650

QUESTION ID

S71-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Uniform model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped, one after the other. P(first is heads and second is tails) is

A. 1/2

B. 1/4

C. 1/8

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

2651

QUESTION ID

S71-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE 4 equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner has 4 equal sections. What is P(one section)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

2652

QUESTION ID
S71-B1-Q5

TYPE
Short answer

CORRECT ANSWER
1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 2

Sample spaces for compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped twice. Which list is the sample space of equally likely outcomes?

A. H, T

B. HH, HT, TT

C. two heads, one head, no heads

D. HH, HT, TH, TT

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

2654

QUESTION ID

S71-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. HH, HT, TH, TT

GRADING NOTE Order matters for equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many outcomes does rolling two dice have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

2655

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

True or false?

When a fair coin is flipped twice, the 3 outcomes "two heads," "one head," and "no heads" are equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

2656

QUESTION ID

S71-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P(one head) = 1/2, while the other two are 1/4 each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many outcomes for two dice give an even sum?

A. 21

B. 12

C. 18

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

2657

QUESTION ID

S71-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 18

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the outcomes of rolling two dice with a sum of 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

2658

QUESTION ID

S71-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(1, 3), (2, 2), (3, 1)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 3

Probability between 0 and 1 and redoing outcomes • Grade 7 • Statistics & Probability • 7.SP.C.5

A fair die is rolled; a roll of 5 or 6 is ignored and the die is rolled again until it shows 1-4. What are the probabilities of the final results 1, 2, 3, and 4?

A. Each becomes 1/5, because one reroll is added.

B. 1 becomes more likely than 4, since it is first.

C. Each stays 1/6, because the die is still fair.

D. They stay equally likely, and each becomes 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

2660

QUESTION ID

S71-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. They stay equally likely, and each becomes 1/4.

GRADING NOTE The four equally likely results are scaled to add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled, and any 5 or 6 is rerolled until the result is 1-4. What is P(final result is 1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

2661

GRADING NOTE Any equivalent form.

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Probabilities of all outcomes in a sample space add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

2662

QUESTION ID

S71-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Total probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An event that can never happen has probability

A. 1

B. −1

C. 1/2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

2663

QUESTION ID

S71-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE Impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Three equally likely outcomes: what is each probability?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

2664

QUESTION ID

S71-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 1

Design fair procedures and count expected trials • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Which procedure chooses one of 5 people fairly using a fair coin?

A. Flip 5 times; the person whose number equals the count of heads is chosen, and 0 heads means flip again.

B. Flip twice (4 equally likely outcomes); one outcome each for persons 1-4, and person 5 also gets HH.

C. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to the people and flip again on the other 3.

D. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to people and give the other 3 to person 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

2666

QUESTION ID
S71-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Flip 3 times (8 equally likely outcomes); assign 5 outcomes to the people and flip again on the other 3.

GRADING NOTE Equal numbers of equally likely outcomes, with the rest redone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three friends choose a driver by flipping a coin twice: HH, HT, and TH each pick a friend, and TT means flip again. What is the probability that a choice is made on the first try?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

2667

QUESTION ID
S71-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A fair procedure must give each person an equal chance, even if it may take several tries.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

2668

QUESTION ID
S71-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Fairness is about probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To choose among 6 people fairly with one die, you

A. roll twice and add

B. assign one face to each person

C. assign 1-3 to person A

D. reroll on 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

2669

QUESTION ID
S71-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. assign one face to each person

GRADING NOTE Six equally likely faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A teacher picks a student from a class of 28 using a random number generator that gives whole numbers 1-30, each equally likely. Which rule is fair?

A. Assign numbers 1-28 to the students, and generate again on 29 or 30.

B. Assign numbers 1-28; 29 and 30 both choose student 1.

C. Assign numbers 1-28; 29 and 30 choose students 1 and 2.

D. Generate one number and divide it by 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

2670

QUESTION ID
S71-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Assign numbers 1-28 to the students, and generate again on 29 or 30.

GRADING NOTE Each student gets one number; extra numbers are redone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 2

Analyze strategies and decisions with probability • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A game costs $1. You roll a fair die and win $4 if you roll a 6, and nothing otherwise. Find the expected net gain per game, in dollars, as a fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

2672

QUESTION ID
S71-F2-Q1

TYPE
Short answer

CORRECT ANSWER
−1/3

GRADING NOTE (1/6)(4) − 1 = −1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A game costs $1. You roll a fair die and win a prize if you roll a 6, and nothing otherwise. With a $4 prize, what is the expected net gain in dollars? What prize makes the expected net gain 0?

Expected net gain ($): _____

Fair prize ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

2673

QUESTION ID
S71-F2-Q2

TYPE
Short answer

CORRECT ANSWER
−1/3; 6

GRADING NOTE Form required: both. (1/6)(4) − 1 = −1/3; (1/6)P − 1 = 0 gives P = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

True or false?

A game can be fair in who is chosen but still have a negative expected value for the player.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

2674

QUESTION ID
S71-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Two meanings of fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

With P(win) = 1/4 and prize \$8 for a \$2 ticket, the expected net gain is

A. \$2

B. −\$2

C. \$0

D. \$6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

2675

QUESTION ID
S71-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. \$0

GRADING NOTE 8/4 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

What is the difference between a fair choice procedure and a fair game?

A. A fair game always has a prize; a fair choice procedure never uses probability at all.

B. A fair choice procedure has expected net gain 0; a fair game gives everyone equal probability.

C. A fair choice procedure gives everyone equal probability; a fair game has expected net gain 0.

D. They mean the same thing: both require each player to have probability 1/2 of winning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

2676

QUESTION ID
S71-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. A fair choice procedure gives everyone equal probability; a fair game has expected net gain 0.

GRADING NOTE Two meanings of fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

SET 72 • HSS-MD.B.7 • STATISTICS, INFERENCE, AND DECISIONS

Analyze decisions and strategies with probabilities

MAIN	Algebra II main question	**HSS-MD.B.7**
BACKWARD 1	Probability of compound independent events	**7.SP.C.8.a**
BACKWARD 2	Two-way tables of counts	**8.SP.A.4**
BACKWARD 3	Weighted averages of outcomes	**6.RP.A.3.c**
FORWARD 1	Investigate how changing conditions changes the decision	**HSS-MD.B.7**
FORWARD 2	Find expected values of random variables	**HSS-MD.A.2**

SET 72 • HSS-MD.B.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A disease affects 1% of 10,000 people. A test detects 95% of people who have it and gives a false positive for 5% of healthy people. Find the number of true positives, the number of false positives, and P(disease | positive) as a decimal rounded to the nearest thousandth.

True positives: _____

False positives: _____

P(disease | positive): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

2679

QUESTION ID

S72-M-Q1

TYPE

Short answer

CORRECT ANSWER

95; 495; 0.161

GRADING NOTE Form required: all three. 0.95(100) = 95; 0.05(9,900) = 495; 95/590 ≈ 0.161.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

A shipment is accepted if 2 randomly tested items are both non-defective. Each item is defective with probability 0.1, independently. What is the probability the shipment is accepted? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

2680

QUESTION ID
S72-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.81

GRADING NOTE 0.9 × 0.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

A $30 warranty covers a repair that costs $200 and is needed with probability 0.1 (otherwise there is no cost). Using expected cost as the decision rule, which is correct?

A. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is worth buying.

B. Expected repair cost $0.9 \times 200 = \$180 > \$30$, so the warranty is worth buying.

C. Expected repair cost $200 - 30 = \$170 > \$30$, so the warranty is worth buying.

D. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is not worth buying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

2681

QUESTION ID
S72-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Expected repair cost $0.1 \times 200 = \$20 < \$30$, so the warranty is not worth buying.

GRADING NOTE On average the warranty costs $10 more than it saves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

Option A pays $10 with probability 0.5 and $0 otherwise; option B pays $50 with probability 0.1 and $0 otherwise. Compare the expected values and the risks.

A. Both have expected value $5; A is riskier, because it pays less.

B. A has expected value $10 and B has $5, so A is better.

C. Both have expected value $5; B is riskier, because it usually pays $0.

D. A has expected value $5 and B has $50, so B is better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

2682

QUESTION ID
S72-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. Both have expected value $5; B is riskier, because it usually pays $0.

GRADING NOTE 0.5(10) = 0.1(50) = 5; B pays $0 90% of the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

A test gives +1 point for a correct answer, −1/4 point for a wrong answer, and 0 for a blank, on 4-choice questions. Using expected score as the decision rule, should a student with no idea of the answer guess randomly or leave it blank?

A. Leave it blank: the expected score of guessing is (1/4)(1) + (3/4)(−1) = −1/2.

B. Guess: the expected score is (1/4)(1) + (3/4)(−1/4) = 1/16, more than 0 for a blank.

C. Leave it blank: a wrong answer loses points, so guessing always lowers the score.

D. Guess: the expected score is (1/4)(1) = 1/4, because wrong answers do not count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

2683

QUESTION ID
S72-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Guess: the expected score is (1/4)(1) + (3/4)(−1/4) = 1/16, more than 0 for a blank.

GRADING NOTE 1/16 > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 1

Probability of compound independent events • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Each item is non-defective with probability 0.9, independently of the others. Find the probability that two items are both non-defective. Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

2685

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.81

GRADING NOTE 0.9 × 0.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins: P(both heads)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

2686

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

True or false?

For independent events, P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

2687

QUESTION ID

S72-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplication rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Each item is non-defective with probability 0.9, independently of the others. The probability that three items are all non-defective is

A. 0.9

B. 0.27

C. 0.729

D. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

2688

QUESTION ID
S72-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 0.729

GRADING NOTE 0.9^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

Each item is defective with probability 0.1, independently of the others. Find the probability that at least one of two items is defective. Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

2689

QUESTION ID

S72-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0.19

GRADING NOTE $1 - 0.9^2 = 0.19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 2

Two-way tables of counts • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Of 10,000 people, 1% have a disease. How many have it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

2691

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 9,900 healthy people, 5% test positive. How many?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

2692

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

495

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

True or false?

A two-way table can show test results for people with and without a disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

2693

QUESTION ID

S72-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rows and columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 95 sick and 495 healthy people test positive, the total positives are

A. 10,000

B. 590

C. 95

D. 495

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

2694

QUESTION ID

S72-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 590

GRADING NOTE Add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

2694

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of positives are sick (95 of 590)? Give a decimal to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

2695

QUESTION ID

S72-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.161

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 3

Weighted averages of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 10% of $200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

2697

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.5 × 10 and 0.1 × 50.

0.5 × 10 = _____

0.1 × 50 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

2698

QUESTION ID

S72-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5; 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1/4 \times 1 + 3/4 \times (-1/4) = 1/4.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q3/5

2699

QUESTION ID

S72-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1/4 - 3/16 = 1/16.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$0.2 \times 30 + 0.8 \times 0$ equals

A. 30

B. 24

C. 0.2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

2700

QUESTION ID

S72-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE Weighted sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.9 × 0 + 0.1 × 200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

2701

QUESTION ID

S72-B3-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 1

Investigate how changing conditions changes the decision • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A test detects 95% of people who have a disease and gives a false positive for 5% of healthy people. In a group of 10,000 where 10% have the disease, find P(disease | positive). Give a decimal rounded to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

2703

QUESTION ID
S72-F1-Q1

TYPE
Short answer

CORRECT ANSWER
0.679

GRADING NOTE 950 true and 0.05(9,000) = 450 false positives; 950/1,400 ≈ 0.679.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A $30 warranty covers a repair that costs $200; the repair is needed with probability 0.2 and otherwise there is no cost. Using expected cost as the decision rule, is the warranty worth buying?

A. Yes: the repair cost of $200 is more than $30.

B. No: the repair probably will not happen, since 0.2 < 0.5.

C. Yes: the expected repair cost is 0.2 × 200 = $40, more than $30.

D. No: the expected repair cost is $40, which is more than $30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

2704

QUESTION ID
S72-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Yes: the expected repair cost is 0.2 × 200 = $40, more than $30.

GRADING NOTE Compare $40 with $30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

True or false?

A positive result is more convincing when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

2705

QUESTION ID
S72-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Compare 0.161 and 0.679.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A test gives +1 point for a correct answer and −1/4 point for a wrong answer on 4-choice questions. A student eliminates one wrong choice and guesses randomly among the 3 remaining choices. What is the expected score for the guess?

A. (1/3)(1) + (2/3)(−1/4) = 1/6

B. (1/3)(1) + (2/3)(−1) = −1/3

C. (1/4)(1) + (3/4)(−1/4) = 1/16

D. (1/3)(1) = 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

2706

QUESTION ID
S72-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A.
(1/3)(1) + (2/3)(−1/4) = 1/6

GRADING NOTE With 3 choices, P(correct) = 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A shipment is accepted only if all n tested items are non-defective. Each item is defective with probability 0.1, independently. What is the smallest whole number n for which the probability of acceptance is below 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

2707

QUESTION ID

S72-F1-Q5

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE $0.9^6 \approx 0.531$ and $0.9^7 \approx 0.478$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 2

Find expected values of random variables • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A random variable X takes values 0, 1, 2 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

2709

QUESTION ID

S72-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A raffle sells 500 tickets at $2 each; one ticket wins a $400 prize. Find the expected net gain of one ticket, in dollars, as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

2710

QUESTION ID
S72-F2-Q2

TYPE
Short answer

CORRECT ANSWER
−1.2

GRADING NOTE 400/500 − 2 = −1.20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q2**
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

True or false?

The expected value is the long-run average of many repetitions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

2711

QUESTION ID
S72-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For a fair die, E(roll) is

A. 3

B. 21

C. 3.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

2712

QUESTION ID

S72-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3.5

GRADING NOTE 21/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A $30 warranty covers a $200 repair that is needed with probability 0.1 (otherwise no cost). Why is deciding whether to buy it an expected-value calculation?

A. The repair cost is a random variable (0 or 200) with expected value $20, compared with the certain $30 price.

B. The repair will cost exactly $20 for certain, so the warranty price is compared with $20.

C. The expected repair cost is 0.1 + 200 = $200.10, which is compared with the $30 price.

D. The warranty price is a random variable whose expected value is $200, compared with the $30 price.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

2713

QUESTION ID
S72-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. The repair cost is a random variable (0 or 200) with expected value $20, compared with the certain $30 price.

GRADING NOTE E(cost) = 0.1(200) + 0.9(0) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSN-CN.A.3 • ADVANCED EXTENSION (+)

Use conjugates to find moduli and quotients of complex numbers

MAIN	Algebra II main question	**HSN-CN.A.3**
BACKWARD 1	Multiply complex numbers	**HSN-CN.A.2**
BACKWARD 2	Distance from the origin	**8.G.B.8**
BACKWARD 3	Powers of i	**HSN-CN.A.1**
FORWARD 1	Prove properties of conjugates and moduli	**HSN-CN.A.3**
FORWARD 2	Write complex numbers in polar form	**HSN-CN.B.4**

SET 73 • HSN-CN.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Give the conjugate of 3 − 4i and find |3 − 4i|.

Conjugate: _____

|3 − 4i| = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

2716

QUESTION ID

S73-M-Q1

TYPE

Short answer

CORRECT ANSWER

3 + 4i; 5

GRADING NOTE Form required: both. √(9 + 16) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q1**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (2 + 3i)/(1 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

2717

QUESTION ID

S73-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 + (5/2)i

GRADING NOTE Form required: a + bi. Multiply by (1 + i)/(1 + i).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (5 + i)/(2 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

2718

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

1 − i

GRADING NOTE Form required: a + bi. (5 + i)(2 − 3i)/13 = (13 − 13i)/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q3**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $|-5 + 12i|$ and compute $(-5 + 12i)(-5 - 12i)$.

$|-5 + 12i| =$ _____

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

2719

QUESTION ID

S73-M-Q4

TYPE

Short answer

CORRECT ANSWER

13; 169

GRADING NOTE Form required: both. $\sqrt{(25 + 144)} = 13$; the product is $25 + 144 = 13^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/(4 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

2720

QUESTION ID

S73-M-Q5

TYPE

Short answer

CORRECT ANSWER

4/25 − (3/25)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q5**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

2720

SET 73 • HSN-CN.A.3

BACKWARD 1

Multiply complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Multiply (2 + 3i)(1 + i).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

2722

QUESTION ID

S73-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−1 + 5i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(3 - 4i)(3 + 4i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

2723

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

True or false?

$(a + bi)(a - bi) = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

2724

QUESTION ID

S73-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-b^2i^2 = b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(5 + i)(2 − 3i) equals

A. 13 + 13i

B. 7 − 13i

C. 10 − 3i

D. 13 − 13i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

2725

QUESTION ID

S73-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13 − 13i

GRADING NOTE 10 − 15i + 2i + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(1 - i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

2726

QUESTION ID

S73-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 2

Distance from the origin • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (3, −4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

2728

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (−5, 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

2729

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The distance from (0, 0) to (a, b) is $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

2730

QUESTION ID

S73-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

2730

ANSWER KEY Reduced copy of the question page, with its answer

The distance from (0, 0) to (1, 1) is

A. 0

B. 1

C. 2

D. √2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

2731

QUESTION ID

S73-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √2

GRADING NOTE √(1 + 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (4, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

2732

QUESTION ID

S73-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 3

Powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

2734

QUESTION ID

S73-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

2734

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-12i \cdot i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

2735

QUESTION ID

S73-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

True or false?

$-3i^2 = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

2736

QUESTION ID

S73-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-3i^2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is real?

A. 3i

B. $(4 + 3i)^2$

C. $(4 + 3i)(4 - 3i)$

D. $4 + 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

2737

QUESTION ID

S73-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(4 + 3i)(4 - 3i)$

GRADING NOTE Conjugate product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 13/13 − 13i/13 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

2738

QUESTION ID

S73-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1 − i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 1

Prove properties of conjugates and moduli • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Which argument proves $z \cdot \bar{z} = |z|^2$ for $z = a + bi$ (a and b real)?

A. $(a + bi)(a - bi) = a^2 + b^2i^2$, which is $|z|^2$ because $|z| = \sqrt{(a^2 + b^2i^2)}$.

B. $(a + bi)(a - bi) = a^2 - b^2$, and $|z|^2 = a^2 - b^2$.

C. $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

D. For $z = 3 + 4i$: $(3 + 4i)(3 - 4i) = 25 = 5^2$, so the statement holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

2740

QUESTION ID

S73-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C.

$(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

GRADING NOTE Multiply, use $i^2 = -1$, and compare with $|z| = \sqrt{(a^2 + b^2)}$. The others forget $i^2 = -1$ (and misstate $|z|$), check one example, or expand with a sign error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Let z = 3 + 4i and w = 1 + i. Find zw, |zw|, and |z||w|, giving the moduli exactly in simplest radical form.

zw = _____

|zw| = _____

|z||w| = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

2741

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−1 + 7i; 5√2; 5√2

GRADING NOTE Form required: all three. zw = 3 + 3i + 4i + 4i² = −1 + 7i; |zw| = √50 = 5√2; |z||w| = 5 · √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

True or false?

The conjugate of a real number is itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

2742

QUESTION ID

S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q3**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $z = 2 - 5i$, then $z + \bar{z}$ is

A. $4 - 10i$

B. 0

C. $-10i$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

2743

QUESTION ID

S73-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\bar{z}/|z|^2$ to write $1/z$ for $z = 1 + 2i$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

2744

QUESTION ID

S73-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1/5 − (2/5)i

GRADING NOTE Form required: $a + bi$. $\bar{z}/|z|^2 = (1 - 2i)/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 2

Write complex numbers in polar form • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 1 + i in polar form r(cos θ + i sin θ) with $0 \le \theta < 2\pi$. Give r and θ exactly.

r = _____

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

2746

QUESTION ID

S73-F2-Q1

TYPE

Short answer

CORRECT ANSWER

√2; π/4

GRADING NOTE Form required: both. r = √(1 + 1); θ = π/4 (first quadrant, equal parts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2(cos(π/2) + i sin(π/2)) in rectangular form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

2747

QUESTION ID

S73-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2i

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

True or false?

The modulus of a complex number is the r in its polar form.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

2748

QUESTION ID
S73-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The polar form of -3 is

A. $3(\cos \pi + i \sin \pi)$

B. $-3(\cos \pi + i \sin \pi)$

C. $3(\cos 0 + i \sin 0)$

D. $3(\cos(\pi/2) + i \sin(\pi/2))$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

2749

QUESTION ID
S73-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 3(cos π + i sin π)

GRADING NOTE Angle π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the complex plane, what do the modulus and the angle of a complex number represent?

A. The modulus is the sum of the parts a + b; the angle is the ratio of the parts b/a.

B. The modulus is the distance from 1; the angle is measured from the positive imaginary axis.

C. The modulus is the real part; the angle is the imaginary part of the number written as a + bi.

D. The modulus is the distance from 0; the angle is the direction measured from the positive real axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

2750

QUESTION ID
S73-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. The modulus is the distance from 0; the angle is the direction measured from the positive real axis.

GRADING NOTE Polar form describes a point by its distance and direction from the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Graph rational functions with zeros and asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Rewrite a rational expression by inspection	**HSA-APR.D.6**
BACKWARD 2	Domain and excluded values	**HSF-IF.B.5**
BACKWARD 3	Sign of a quotient on intervals	**7.NS.A.2**
FORWARD 1	Graph rational functions with slant asymptotes	**HSF-IF.C.7.d**
FORWARD 2	Interpret a rational model's features in context	**HSF-IF.B.4**

SET 74 • HSF-IF.C.7.d

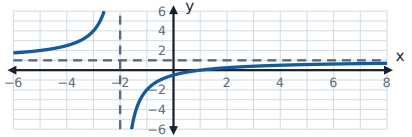
MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

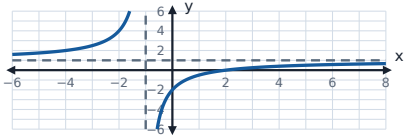
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = (x + 1)/(x − 2), with its asymptotes dashed?

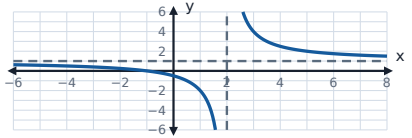
A



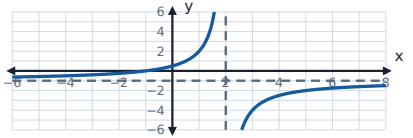
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

2753

QUESTION ID

S74-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Zero −1, y-intercept −1/2, vertical asymptote x = 2, horizontal asymptote y = 1 (ratio of leading coefficients); $y \to +\infty$ as $x \to 2^+$ and $y \to -\infty$ as $x \to 2^-$. The others show (x − 1)/(x + 2) (same y-intercept), (x − 2)/(x + 1), and the reflection.

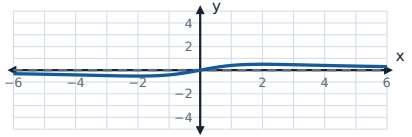
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

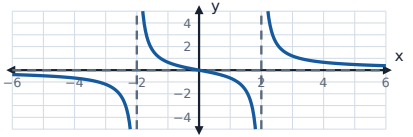
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2x/(x^2 - 4)$, with its asymptotes dashed?

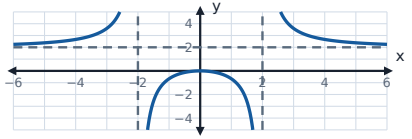
A



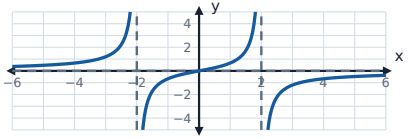
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

2754

QUESTION ID

S74-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Zero and y-intercept 0; vertical asymptotes $x = \pm 2$ (zeros of $x^2 - 4$); horizontal asymptote $y = 0$ (degree of numerator < degree of denominator); e.g. $y(3) = 6/5 > 0$ and $y(1) = -2/3 < 0$. The others show $2x/(x^2 + 4)$, $2x^2/(x^2 - 4)$, and the reflection.

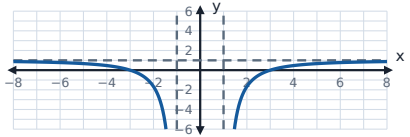
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

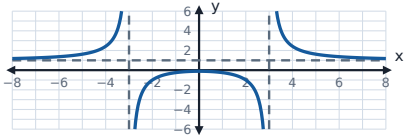
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 - 1)/(x^2 - 9)$, with its asymptotes dashed?

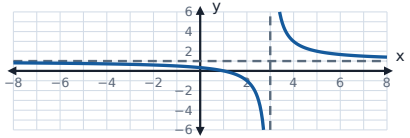
A



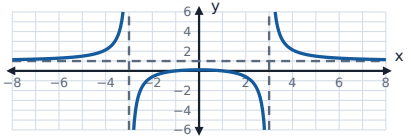
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

2755

QUESTION ID

S74-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Zeros ±1, y-intercept 1/9, vertical asymptotes $x = \pm 3$, horizontal asymptote $y = 1$; the outer branches are above $y = 1$. The others show the reciprocal $(x^2 - 9)/(x^2 - 1)$, $(x^2 + 1)/(x^2 - 9)$, and $(x - 1)/(x - 3)$.

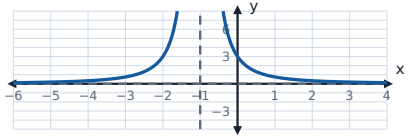
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

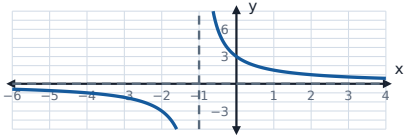
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 3/(x + 1)^2$, with its asymptotes dashed?

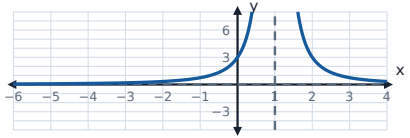
A



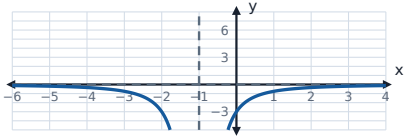
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

2756

QUESTION ID

S74-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE No zeros; y-intercept 3; vertical asymptote $x = -1$ with $y \to +\infty$ on both sides (the square is positive); horizontal asymptote $y = 0$. The others show $3/(x + 1)$, $3/(x - 1)^2$ (same y-intercept), and $-3/(x + 1)^2$.

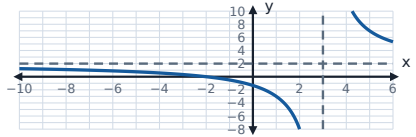
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

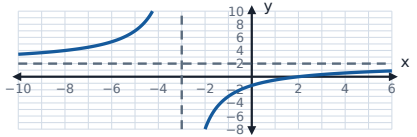
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = (2x − 4)/(x + 3), with its asymptotes dashed?

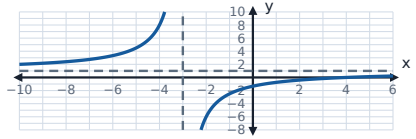
A



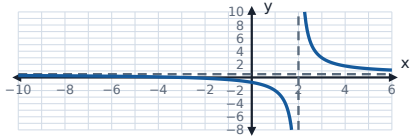
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

2757

QUESTION ID

S74-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Zero 2, y-intercept −4/3, vertical asymptote x = −3, horizontal asymptote y = 2/1 = 2. The others show (2x + 4)/(x − 3) and (x − 4)/(x + 3) (both with the same y-intercept) and the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 1

Rewrite a rational expression by inspection • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite (x + 1)/(x − 2) as 1 + r/(x − 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

2759

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 + 3/(x − 2)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite (2x − 4)/(x + 3) as 2 + r/(x + 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

2760

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 − 10/(x + 3)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

True or false?

The constant q in q + r/(x − a) is the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

2761

QUESTION ID

S74-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE r/(x − a) → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 1)/(x^2 - 9)$ equals

A. $1 + 8/(x^2 - 9)$

B. $1 - 8/(x^2 - 9)$

C. $1 + 1/(x^2 - 9)$

D. $1/9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

2762

QUESTION ID

S74-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $1 + 8/(x^2 - 9)$

GRADING NOTE $x^2 - 1 = (x^2 - 9) + 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor the denominator of $2x/(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

2763

QUESTION ID

S74-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)$

GRADING NOTE Form required: factored. The vertical asymptotes are at the zeros of these factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 2

Domain and excluded values • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $(x + 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

2765

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \neq 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

2765

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $3/(x + 1)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

2766

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \neq -1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

True or false?

A rational function is undefined where its denominator is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

2767

QUESTION ID

S74-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division by 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$2x/(x^2 - 4)$ is undefined at

A. $x = \pm 2$

B. $x = 4$

C. $x = 2$ only

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

2768

QUESTION ID

S74-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x = \pm 2$

GRADING NOTE $x^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the excluded values of $(x^2 - 1)/(x^2 - 9)$, separated by commas.

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

2769

QUESTION ID

S74-B2-Q5

TYPE

Short answer

CORRECT ANSWER

{−3, 3}

GRADING NOTE Form required: both. $x^2 - 9 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 3

Sign of a quotient on intervals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is (x + 1)/(x − 2) positive or negative at x = 0? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

2771

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE 1/(−2) < 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

2771

ANSWER KEY Reduced copy of the question page, with its answer

Is $2x/(x^2 - 4)$ positive or negative at $x = 3$? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

2772

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE $6/5 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A quotient of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

2773

QUESTION ID

S74-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example $(-6)/(-2) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

At x = 2.1, (x + 1)/(x − 2) is

A. large and negative

B. undefined

C. zero

D. large and positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

2774

QUESTION ID

S74-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. large and positive

GRADING NOTE 3.1/0.1 = 31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$3/(x + 1)^2$ is negative for some value of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

2775

QUESTION ID

S74-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A positive number over a positive square is positive (and x = −1 is excluded).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSF-IF.C.7.d

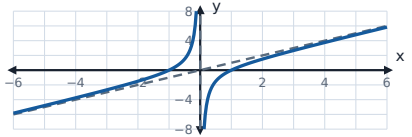
FORWARD 1

Graph rational functions with slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

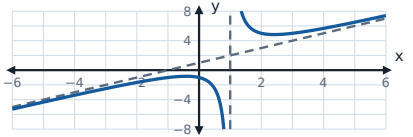
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 + 1)/x$, with its asymptotes dashed?

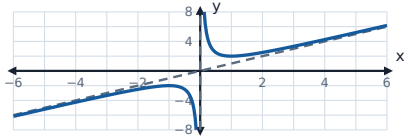
A



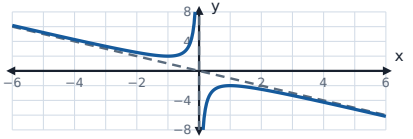
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

2777

QUESTION ID

S74-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $(x^2 + 1)/x = x + 1/x$: vertical asymptote $x = 0$, slant asymptote $y = x$, no zeros, no y-intercept, through $(1, 2)$ and $(-1, -2)$. The others show $(x^2 - 1)/x$, $(x^2 + 1)/(x - 1)$, and $-(x^2 + 1)/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slant asymptote of $y = (x^2 - x - 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

2778

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$y = x + 1$

GRADING NOTE Form required: equation. Quotient x + 1, remainder 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

True or false?

A rational function can cross its horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

2779

QUESTION ID

S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example $y = x/(x^2 + 1)$ crosses $y = 0$ at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (3x^2 + 1)/(x^2 + 4)$ has horizontal asymptote

A. none

B. y = 3

C. y = 0

D. y = 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

2780

QUESTION ID

S74-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 3

GRADING NOTE Ratio of leading coefficients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

Which rational function has vertical asymptote x = 1, horizontal asymptote y = 2, and a zero at x = 3?

A. f(x) = (x − 3)/(2(x − 1))

B. f(x) = 2(x − 3)/(x − 1)

C. f(x) = 2(x − 1)/(x − 3)

D. f(x) = 2(x + 3)/(x + 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

2781

QUESTION ID

S74-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. f(x) = 2(x − 3)/(x − 1)

GRADING NOTE Zero from the numerator factor x − 3; asymptote from x − 1; ratio of leading coefficients 2/1. The others give y = 1/2, swap the roles, or change the signs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

FORWARD 2

Interpret a rational model's features in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The average cost per item is $A(x) = (500 + 4x)/x$ dollars for x items. What do the vertical asymptote $x = 0$ and the horizontal asymptote $y = 4$ mean in context?

A. x = 0: the fixed cost of production is $0; y = 4: no more than 4 items can be made at this average cost per item.

B. x = 0: the average cost grows without bound as the number of items nears 0; y = 4: the average cost approaches $4 per item as x grows.

C. x = 0: the first item is free to make; y = 4: the total cost of all of the items approaches $4 as the number of items grows.

D. x = 0: the average cost is exactly $500 for very few items; y = 4: the average cost is never more than $4 per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

2783

QUESTION ID
S74-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. x = 0: the average cost grows without bound as the number of items nears 0; y = 4: the average cost approaches $4 per item as x grows.

GRADING NOTE 500/x grows without bound near 0 and shrinks toward 0 as x grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ for $t \geq 0$ hours. What does its horizontal asymptote mean?

A. y = 0: the concentration is 0 at every time after t = 2.

B. y = 20: the concentration approaches 20 as time goes on.

C. y = 0: the drug never enters the blood.

D. y = 0: the concentration approaches 0 as time goes on.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

2784

QUESTION ID
S74-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. y = 0: the concentration approaches 0 as time goes on.

GRADING NOTE The degree of the denominator is greater, so C → 0; C is positive for t > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

True or false?

For a context, only the part of a rational graph in the context domain is meaningful.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

2785

QUESTION ID
S74-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In A(x) = (500 + 4x)/x, as x grows the average cost

A. increases

B. is constant

C. approaches 0

D. decreases toward 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

2786

QUESTION ID

S74-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. decreases toward 4

GRADING NOTE 500/x → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The average cost per item is A(x) = (500 + 4x)/x dollars for x items. Find the number of items for which the average cost is $5.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

2787

QUESTION ID
S74-F2-Q5

TYPE
Short answer

CORRECT ANSWER
500

GRADING NOTE 500 + 4x = 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Distinguish holes from vertical asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 2	Simplify rational expressions and keep exclusions	**HSA-APR.D.7**
BACKWARD 3	Evaluate the simplified expression	**HSF-IF.A.2**
FORWARD 1	Construct and graph rational functions with holes	**HSF-IF.C.7.d**
FORWARD 2	Relate holes to extraneous solutions	**HSA-REI.A.2**

SET 75 • HSF-IF.C.7.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x^2 - 4)/(x - 2)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

2790

QUESTION ID
S75-M-Q1

TYPE
Short answer

CORRECT ANSWER
(2, 4); none

GRADING NOTE Form required: both. f(x) = x + 2 for x ≠ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 3)/(x^2 - 9)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

2791

QUESTION ID
S75-M-Q2

TYPE
Short answer

CORRECT ANSWER
(3, 1/6); −3

GRADING NOTE Form required: both. $f(x) = 1/(x + 3)$ for $x \neq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x^2 + x - 6)/(x^2 - x - 2)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

2792

QUESTION ID

S75-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 5/3); −1

GRADING NOTE Form required: both. $(x + 3)(x - 2)/((x - 2)(x + 1)) = (x + 3)/(x + 1)$ for $x \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x + 1)/(x^2 - 1)$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

2793

QUESTION ID

S75-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−1, −1/2); 1

GRADING NOTE Form required: both. $f(x) = 1/(x - 1)$ for $x \neq -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x(x - 1)/(x(x + 2))$ has exactly one hole. Give the hole as a point (x, y), and give every vertical asymptote as an x-value (enter none if there is none).

Hole: _____

Vertical asymptote: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

2794

QUESTION ID

S75-M-Q5

TYPE

Short answer

CORRECT ANSWER

(0, −1/2); −2

GRADING NOTE Form required: both. $f(x) = (x - 1)/(x + 2)$ for $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 1

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + x - 6$ and $x^2 - x - 2$.

$x^2 + x - 6 =$ _____

$x^2 - x - 2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

2796

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x − 2);
(x − 2)(x + 1)

GRADING NOTE Form required: both factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

2797

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 - 1 = (x - 1)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

2798

QUESTION ID

S75-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 1)^2 = x^2 - 2x + 1$; $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

2799

QUESTION ID

S75-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (x − 2)(x + 2)

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

2800

QUESTION ID

S75-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x(x + 2)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

2800

SET 75 • HSF-IF.C.7.d

BACKWARD 2

Simplify rational expressions and keep exclusions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (x − 3)/((x − 3)(x + 3)) and list the excluded values of the original expression, separated by commas.

Simplified: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

2802

QUESTION ID
S75-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/(x + 3); {−3, 3}

GRADING NOTE Form required: both. The canceled factor still excludes x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (x + 1)/((x − 1)(x + 1)) and list the excluded values of the original expression, separated by commas.

Simplified: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

2803

QUESTION ID
S75-B2-Q2

TYPE
Short answer

CORRECT ANSWER
1/(x − 1); {−1, 1}

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-B2-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

After simplifying, the original excluded values are still excluded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

2804

QUESTION ID
S75-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Domain kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x(x - 1)/(x(x + 2))$ simplifies to

A. $-1/2$

B. $(x - 1)/(x + 2)$ for all x

C. $x - 1$

D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

2805

QUESTION ID
S75-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

GRADING NOTE Cancel x but keep $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2 - 4)/(x - 2)$ and give the excluded value of the original expression.

Simplified: _____

Excluded value: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

2806

QUESTION ID
S75-B2-Q5

TYPE
Short answer

CORRECT ANSWER
x + 2; 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-B2-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 3

Evaluate the simplified expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1/(x + 3) at x = 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

2808

QUESTION ID

S75-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate (x + 3)/(x + 1) at x = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

2809

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

True or false?

The y-coordinate of a hole is the value of the simplified expression there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

2810

QUESTION ID

S75-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Limit value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 1)/(x + 2)$ at $x = 0$ is

A. 1/2

B. −1/2

C. undefined

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

2811

QUESTION ID

S75-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1/2

GRADING NOTE −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

2811

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1/(x − 1) at x = −1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

2812

QUESTION ID

S75-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 1

Construct and graph rational functions with holes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

Which rational function has a hole at x = 1, a vertical asymptote x = −3, and horizontal asymptote y = 0?

A. f(x) = (x − 1)/((x − 1)(x + 3))

B. f(x) = (x − 1)(x + 3)/(x − 1)

C. f(x) = (x − 1)/(x + 3)

D. f(x) = (x + 3)/((x − 1)(x + 3))

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

2814

QUESTION ID

S75-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. f(x) = (x − 1)/((x − 1)(x + 3))

GRADING NOTE The shared factor x − 1 gives the hole; x + 3 remains below; the denominator has the higher degree. The others have no hole, swap the hole and asymptote, or have no asymptote.

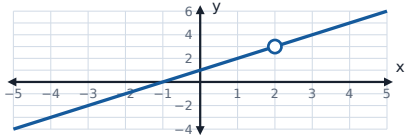
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

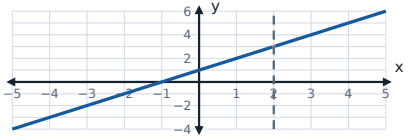
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 - x - 2)/(x - 2)$?

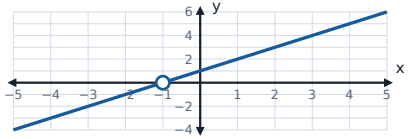
A



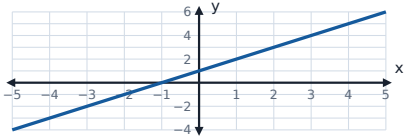
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

2815

QUESTION ID

S75-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $x^2 - x - 2 = (x - 2)(x + 1)$, so $y = x + 1$ for $x \neq 2$: the line with a hole at (2, 3). The factor cancels, so there is no asymptote at $x = 2$, and $x = -1$ is a zero, not an excluded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

True or false?

A rational function can have both a hole and a vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

2816

QUESTION ID

S75-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Different factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

$f(x) = (x - 4)^2/(x - 4)$ has

A. no special features

B. a hole at (4, 4)

C. a vertical asymptote $x = 4$

D. a hole at (4, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q4/5

2817

QUESTION ID

S75-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a hole at (4, 0)

GRADING NOTE Simplifies to $x - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = (x - 3)/(x^2 - 9)$, why is $x = -3$ a vertical asymptote but $x = 3$ a hole?

A. Negative excluded values always give asymptotes, and positive excluded values always give holes in a graph.

B. Both $x - 3$ and $x + 3$ cancel, so $x = -3$ and $x = 3$ are both holes, and the asymptote comes from x^2 alone.

C. After canceling $x - 3$, the factor $x + 3$ remains in the denominator, but $x - 3$ does not; near 3 the values approach 1/6.

D. The numerator is 0 at $x = -3$, so the graph has a vertical asymptote at -3, and a hole is at the other zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

2818

QUESTION ID
S75-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. After canceling $x - 3$, the factor $x + 3$ remains in the denominator, but $x - 3$ does not; near 3 the values approach 1/6.

GRADING NOTE $(x - 3)/((x - 3)(x + 3)) = 1/(x + 3)$ for $x \neq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 2

Relate holes to extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $(x^2 - 4)/(x - 2) = 4$.

A. $x = 4$, because the hole of the graph is at the height $y = 4$.

B. $x = 2$, because it satisfies the simplified equation $x + 2 = 4$.

C. Every $x \neq 2$, because the function equals 4 at every point of its domain.

D. No solution: $x + 2 = 4$ gives $x = 2$, which is excluded (the hole at (2, 4)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

2820

QUESTION ID
S75-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. No solution: $x + 2 = 4$ gives $x = 2$, which is excluded (the hole at (2, 4)).

GRADING NOTE The simplified equation has the only candidate $x = 2$, which is not in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 3)/(x^2 - 9) = 1/6$.

A. $x = 3$ or $x = -3$: both make $x^2 - 9 = 0$, so both solve it.

B. $x = -3$: then $1/(x + 3) = 1/6$ after simplifying the left side.

C. No solution: $1/(x + 3) = 1/6$ gives $x = 3$, which is excluded.

D. $x = 3$: it satisfies $1/(x + 3) = 1/6$, the simplified equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

2821

QUESTION ID
S75-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. No solution: 1/(x + 3) = 1/6 gives x = 3, which is excluded.

GRADING NOTE The only candidate is the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of f(x) = c can occur exactly at a hole whose height is c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

2822

QUESTION ID

S75-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Graph has no point there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)/(x^2 - 1) = -1/2$ has

A. no solution

B. $x = -1$

C. $x = 0$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q4/5

2823

QUESTION ID

S75-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. no solution

GRADING NOTE The only candidate is the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x^2 + x - 6)/(x^2 - x - 2) = 2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

2824

QUESTION ID

S75-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. (x + 3)/(x + 1) = 2 gives x + 3 = 2x + 2, so x = 1, which is not excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c • ADVANCED EXTENSION (+)

Compose functions and interpret compositions with domains

MAIN	Algebra II main question	**HSF-BF.A.1.c**
BACKWARD 1	Build combined models	**HSF-BF.A.1.b**
BACKWARD 2	Evaluate a function at an expression	**HSF-IF.A.2**
BACKWARD 3	Domains of radical, rational, and log functions	**HSF-IF.B.5**
FORWARD 1	Decompose functions into compositions	**HSF-BF.A.1.c**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 76 • HSF-BF.A.1.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$ and $g(x) = x - 4$. Which gives $f(g(x))$ and $g(f(x))$ with the domain of each?

A. $f(g(x)) = \sqrt{x - 4}$, $x \geq 0$; $g(f(x)) = \sqrt{(x - 4)}$, $x \geq 4$

B. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 4$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 0$

C. $f(g(x)) = \sqrt{(x - 4)}$, all real x; $g(f(x)) = \sqrt{x} - 4$, all real x

D. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 0$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

2827

QUESTION ID

S76-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $f(g(x)) = \sqrt{(x - 4)}$, $x \geq 4$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 0$

GRADING NOTE The input of a square root must be nonnegative: $x - 4 \geq 0$ for $f(g(x))$, $x \geq 0$ for $g(f(x))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The temperature is T(h) = 20 − 6.5h °C at height h km, and a balloon's height is h(t) = 0.3t km after t minutes. Write T(h(t)) = a + bt, and find the temperature at t = 10.

a = _____

b = _____

Temperature at t = 10 (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

2828

QUESTION ID

S76-M-Q2

TYPE

Short answer

CORRECT ANSWER

20; −1.95; 0.5

GRADING NOTE Form required: all three. T(h(t)) = 20 − 6.5(0.3t) = 20 − 1.95t; T = 20 − 19.5 = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 1/(x - 1)$ and $g(x) = x^2$. Write $f(g(x))$ and list the real numbers excluded from its domain, separated by commas. Use ^ for exponents.

$f(g(x)) =$ _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

2829

QUESTION ID

S76-M-Q3

TYPE

Short answer

CORRECT ANSWER

$1/(x^2 - 1)$; $\{-1, 1\}$

GRADING NOTE Form required: both. $x^2 - 1 = 0$ when $x = \pm 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

A dollar amount d converts to euros by e(d) = 0.9d, and a store's price with tax in euros is p(e) = 1.2e. Then p(e(d)) = kd. Find k and the price in euros for d = 50.

k = _____

p(e(50)) (euros): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

2830

QUESTION ID
S76-M-Q4

TYPE
Short answer

CORRECT ANSWER
1.08; 54

GRADING NOTE Form required: both. 1.2(0.9d) = 1.08d; 1.08 · 50 = 54.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

$f(x) = \log_2 x$ and $g(x) = x^2 - 9$. Which gives $f(g(x))$ and its domain?

A. $f(g(x)) = \log_2(x^2 - 9)$; domain $x < -3$ or $x > 3$

B. $f(g(x)) = (\log_2 x)^2 - 9$; domain $x > 0$

C. $f(g(x)) = \log_2(x^2) - 9$; domain $x \neq 0$

D. $f(g(x)) = \log_2(x^2 - 9)$; domain $x > 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

2831

QUESTION ID
S76-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A.
$f(g(x)) = \log_2(x^2 - 9)$; domain $x < -3$ or $x > 3$

GRADING NOTE $x^2 - 9 > 0$ on both sides. The others compose in the wrong place, miss $x < -3$, or compose in the other order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 1

Build combined models • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) + g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

2833

QUESTION ID

S76-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) \cdot g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

2834

QUESTION ID

S76-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

True or false?

f(g(x)) is a different combination from f(x) · g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

2835

QUESTION ID

S76-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Composition vs product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = 2x and g(x) = x + 3, f(x) − g(x) is

A. x − 3

B. $2x^2 + 6x$

C. x + 3

D. 3x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

2836

QUESTION ID
S76-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. x − 3

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

Which operation combines T and h in T(h(t))?

A. division

B. addition

C. composition

D. multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

2837

QUESTION ID

S76-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. composition

GRADING NOTE The output of h is the input of T.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 2

Evaluate a function at an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √x. Find f(x − 4).

A. √x − √4

B. √(x) · (−4)

C. √x − 4

D. √(x − 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

2839

QUESTION ID

S76-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. √(x − 4)

GRADING NOTE Replace x by x − 4 inside the root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

T(h) = 20 − 6.5h. Find T(0.3t) in the form a + bt. Give a and b.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

2840

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20; −1.95

GRADING NOTE Form required: both. 6.5 · 0.3 = 1.95.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

2840

True or false?

If $f(x) = 1/(x - 1)$, then $f(x^2) = 1/(x^2 - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

2841

QUESTION ID

S76-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $g(x) = x^2 - 9$, then $g(3)$ is

A. -3

B. -6

C. 9

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

2842

QUESTION ID

S76-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $9 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

p(e) = 1.2e. Find p(45).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

2843

QUESTION ID

S76-B2-Q5

TYPE

Short answer

CORRECT ANSWER

54

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 3

Domains of radical, rational, and log functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{(x - 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

2845

QUESTION ID

S76-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log_2 u$ in terms of u.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

2846

QUESTION ID

S76-B3-Q2

TYPE

Short answer

CORRECT ANSWER

u > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

True or false?

$1/(x^2 - 1)$ is undefined only at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

2847

QUESTION ID

S76-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 1 = 0$ at both $x = 1$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 9 > 0$ when

A. $x < -3$ or $x > 3$

B. $-3 < x < 3$

C. all x

D. $x > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

2848

QUESTION ID

S76-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x < -3$ or $x > 3$

GRADING NOTE Outside the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{x} - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

2849

QUESTION ID

S76-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 1

Decompose functions into compositions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Which pair writes $h(x) = (3x + 1)^5$ as $f(g(x))$?

A. $g(x) = 3x + 1$, $f(x) = x^5$

B. $g(x) = 3x + 1$, $f(x) = 5x$

C. $g(x) = 3x$, $f(x) = x^5 + 1$

D. $g(x) = x^5$, $f(x) = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

2851

QUESTION ID

S76-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $g(x) = 3x + 1$, $f(x) = x^5$

GRADING NOTE The inner function is $3x + 1$ and the outer raises to the 5th power. The others reverse the order, split the inner function, or multiply by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair writes $h(x) = 2^{x^2}$ as $f(g(x))$?

A. $g(x) = 2x$, $f(x) = x^2$

B. $g(x) = x^2$, $f(x) = 2^x$

C. $g(x) = x^2$, $f(x) = 2x$

D. $g(x) = 2^x$, $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

2852

QUESTION ID

S76-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $g(x) = x^2$, $f(x) = 2^x$

GRADING NOTE $f(g(x)) = 2^{x^2}$. The others give $(2^x)^2$, $(2x)^2$, or $2x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$h(x) = (x + 1)^2$ can be written as $f(g(x))$ in more than one way, for example with $g(x) = x + 1$ and $f(x) = x^2$, or with $g(x) = x^2 + 2x$ and $f(x) = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

2853

QUESTION ID
S76-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both give $x^2 + 2x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(x) = x + 1$, $g(x) = 2x$, $k(x) = x^2$, then $f(g(k(3)))$ is

A. 37

B. 19

C. 64

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

2854

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 19

GRADING NOTE $k(3) = 9$, $g = 18$, $f = 19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

Which context describes a composition V(r(t))?

A. The time t depends on the volume V, and the volume depends on the radius r.

B. The volume V of a balloon depends on its radius r, and the radius depends on time t.

C. The volume V of a balloon is multiplied by its radius r at each time t.

D. The volume V and the radius r of a balloon are added together at each time t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

2855

QUESTION ID

S76-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The volume V of a balloon depends on its radius r, and the radius depends on time t.

GRADING NOTE t → r → V: the output of r is the input of V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Which work shows that f(x) = 2x + 3 and g(x) = (x − 3)/2 are inverses?

A. f(0) = 3 and g(0) = −3/2, which are opposite in sign

B. f(x) · g(x) = (2x + 3)(x − 3)/2 = x

C. f(g(x)) = 2((x − 3)/2) + 3 = x + 6 and g(f(x)) = x

D. f(g(x)) = 2((x − 3)/2) + 3 = x and g(f(x)) = ((2x + 3) − 3)/2 = x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

2857

QUESTION ID
S76-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. f(g(x)) = 2((x − 3)/2) + 3 = x and g(f(x)) = ((2x + 3) − 3)/2 = x

GRADING NOTE Both compositions must equal x. The others make an arithmetic error, multiply instead of composing, or compare single values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute f(g(x)) for $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

2858

QUESTION ID

S76-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

True or false?

If f(g(x)) = x for all x in g's domain and g(f(x)) = x for all x in f's domain, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

2859

QUESTION ID

S76-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = x + 5 and g(x) = x − 5, f(g(2)) is

A. 2

B. −3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

2860

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE Inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is composition the natural test for inverse functions?

A. Composition adds the two functions, and inverses must add to 0 for every input x.

B. Composition reverses the graph across the x-axis, which is where an inverse must lie.

C. An inverse undoes the function, so applying one after the other must return the original input.

D. Two functions are inverses exactly when their product f(x) · g(x) is equal to 1 for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

2861

QUESTION ID

S76-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. An inverse undoes the function, so applying one after the other must return the original input.

GRADING NOTE f(g(x)) = x and g(f(x)) = x. The others describe adding, reflecting over the x-axis, or multiplying, none of which is the inverse relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b • ADVANCED EXTENSION (+)

Verify inverses by composition on the stated domains

MAIN	Algebra II main question	**HSF-BF.B.4.b**
BACKWARD 1	Find inverses algebraically	**HSF-BF.B.4.a**
BACKWARD 2	Compose functions	**HSF-BF.A.1.c**
BACKWARD 3	Square roots of squares	**HSN-RN.A.2**
FORWARD 1	Show that proposed inverses fail	**HSF-BF.B.4.b**
FORWARD 2	Read inverse values from tables and graphs	**HSF-BF.B.4.c**

SET 77 • HSF-BF.B.4.b

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = 1$, so f and g are inverses.

B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

C. $f(1) = 5$ and $g(5) = 1$, so f and g are inverses.

D. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ for all real x, and one composition is enough by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

2864

QUESTION ID
S77-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

GRADING NOTE Both compositions must simplify to x. The others check one composition only, one input only, or multiply the functions instead of composing them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = x^3 - 1$ and $g(x) = \sqrt[3]{(x + 1)}$ are inverses?

A. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$, but only for $x \geq 0$, so f and g are inverses only when $x \geq 0$.

B. $g(f(x)) = \sqrt[3]{(x^3 - 1)} + 1 = x$, so f and g are inverses.

C. $f(2) = 7$ and $g(7) = 2$, so f and g are inverses.

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

2865

QUESTION ID

S77-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

GRADING NOTE Both compositions simplify to x for every real x (cube roots are defined for negatives). The others add a false domain limit, check one input, or compose incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverses, with the correct domains?

A. $f(g(x)) = 2 \cdot \log_2 x = x$ for $x > 0$.

B. $f(g(x)) = 2^{\log_2 x} = x$ and $g(f(x)) = \log_2(2^x) = x$, both for every real number x.

C. $f(3) = 8$ and $g(8) = 3$, so f and g are inverses for all x.

D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

2866

QUESTION ID
S77-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

GRADING NOTE $\log_2 x$ is defined only for $x > 0$, so $f(g(x)) = x$ holds for $x > 0$; $g(f(x)) = x$ for every real x. The others ignore the domain of $\log_2$, multiply instead of composing, or check one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 1/(x - 2)$ $(x \neq 2)$ and $g(x) = 1/x + 2$ $(x \neq 0)$ are inverses?

A. $f(3) = 1$ and $g(1) = 3$, so f and g are inverses.

B. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$, so f and g are inverses for all real x with no excluded values needed.

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

D. $g(f(x)) = 1/(x - 2) + 2 = (2x - 3)/(x - 2)$, so they are not inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

2867

QUESTION ID

S77-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

GRADING NOTE Both compositions simplify to x on the stated domains. The others drop the second composition and the domains, compute $f(x) + 2$ instead of $g(f(x))$, or check one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ with domain $x \geq 0$ and $g(x) = \sqrt{x}$. Which work verifies that they are inverses and explains why the domain restriction is needed?

A. The restriction is needed because $\sqrt{x}$ is undefined for $x > 0$.

B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

C. $f(g(x)) = (\sqrt{x})^2 = x$ and $g(f(x)) = \sqrt{(x^2)} = x$ for every real number x, so the two functions are inverses and no domain restriction is needed.

D. $g(f(x)) = \sqrt{(x^2)} = \pm x$, and the restriction chooses the + sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

2868

QUESTION ID
S77-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

GRADING NOTE $\sqrt{(x^2)} = |x|$, which equals x only when $x \geq 0$ (for example $\sqrt{((-3)^2)} = 3$). The others ignore $|x|$, state a false fact about $\sqrt{x}$, or treat $\sqrt{}$ as giving two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 1

Find inverses algebraically • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of f(x) = 2x + 3.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

2870

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = x^3 - 1$.

A. $f^{-1}(x) = \sqrt[3]{(x + 1)}$

B. $f^{-1}(x) = \sqrt[3]{x} + 1$

C. $f^{-1}(x) = \sqrt[3]{(x - 1)}$

D. $f^{-1}(x) = 1/(x^3 - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

2871

QUESTION ID

S77-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $f^{-1}(x) = \sqrt[3]{(x + 1)}$

GRADING NOTE $y + 1 = x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The inverse of 2^x is $\log_2 x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

2872

QUESTION ID

S77-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The inverse of f(x) = 1/(x − 2) is

A. 1/x + 2

B. 1/(x + 2)

C. 2 − 1/x

D. x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

2873

QUESTION ID
S77-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 1/x + 2

GRADING NOTE Solve y = 1/(x − 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = x^2$ for $x \geq 0$.

A. $f^{-1}(x) = \pm\sqrt{x}$

B. $f^{-1}(x) = \sqrt{x}$

C. $f^{-1}(x) = x/2$

D. $f^{-1}(x) = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

2874

QUESTION ID

S77-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. $f^{-1}(x) = \sqrt{x}$

GRADING NOTE The nonnegative root undoes squaring on $x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 2

Compose functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 2x + 3 and g(x) = (x − 3)/2. Compute f(g(5)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

2876

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x + 3$ and $g(x) = (x - 3)/2$. Compute $g(f(5))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

2877

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-B2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

True or false?

f(g(x)) means apply g first, then f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

2878

QUESTION ID
S77-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Inside out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^3$ and $g(x) = x + 1$, $f(g(x))$ is

A. $(x + 1)^3$

B. $x^3 + 1$

C. $x + 1$

D. $3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

2879

QUESTION ID

S77-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $(x + 1)^3$

GRADING NOTE Substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $\log_2(2^5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

2880

QUESTION ID
S77-B2-Q5

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 3

Square roots of squares • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt{((-3)^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

2882

QUESTION ID

S77-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

2883

QUESTION ID
S77-B3-Q2

TYPE
Short answer

CORRECT ANSWER
7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

2884

QUESTION ID

S77-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{(x^2)}$ equals

A. $|x|$

B. x^2

C. $-x$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

2885

QUESTION ID
S77-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $|x|$

GRADING NOTE Principal root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $\sqrt{(x^2)} = x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

2886

QUESTION ID

S77-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 1

Show that proposed inverses fail • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Which work shows by composition that $f(x) = 3x - 1$ and $g(x) = (x - 1)/3$ are not inverses?

A. $f(0) = -1$ and $g(0) = -1/3$ are different, so they are not inverses

B. $f(g(x)) = 3((x - 1)/3) - 1 = x$, so they are inverses

C. $g(f(x)) = ((3x - 1) - 1)/3 = x$, so they are inverses

D. $f(g(x)) = 3((x - 1)/3) - 1 = x - 2$, which is not x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

2888

QUESTION ID

S77-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. $f(g(x)) = 3((x - 1)/3) - 1 = x - 2$, which is not x

GRADING NOTE One composition that is not x is enough. The others make arithmetic errors or compare f and g at the same input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ (all real x) and $g(x) = \sqrt{x}$. Which composition fails to equal x, and for which x?

A. $g(f(x)) = \sqrt{(x^2)} = |x|$, which is not x for $x < 0$

B. $f(g(x)) = (\sqrt{x})^2$ fails for every $x > 0$

C. $g(f(x)) = x$ for every real x

D. $f(g(x)) = |x|$, which is not x for $x < 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

2889

QUESTION ID
S77-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. $g(f(x)) = \sqrt{(x^2)} = |x|$, which is not x for $x < 0$

GRADING NOTE $\sqrt{(x^2)} = |x|$. f(g(x)) is defined only for $x \geq 0$, where it equals x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

True or false?

Checking f(g(x)) = x alone is always enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

2890

QUESTION ID
S77-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both compositions must be checked (with domains).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = e^x$ and $g(x) = \ln x$, $g(f(x)) =$

A. $|x|$

B. x only for $x > 0$

C. x for all real x

D. $e^{\ln x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

2891

QUESTION ID

S77-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x for all real x

GRADING NOTE $\ln(e^x) = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Fix the domain of $f(x) = (x - 1)^2$ so that $g(x) = 1 + \sqrt{x}$ is its inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

2892

QUESTION ID

S77-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 2

Read inverse values from tables and graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(7)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

2894

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(f^{-1}(15))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

2895

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(3) = 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

True or false?

If (a, b) is on the graph of f, then (b, a) is on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

2896

QUESTION ID

S77-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(4) = 9$, then $f^{-1}(9)$ is

A. 1/4

B. 4

C. 13

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

2897

QUESTION ID

S77-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(1) = 3 and f(2) = 7. How do these values give $f(f^{-1}(7)) = 7$?

A. $f^{-1}(7) = 1/7$ because an inverse is a reciprocal, and $f(1/7) = 7$, so $f(f^{-1}(7)) = 7$.

B. $f^{-1}(7) = 7 - 3 = 4$ by subtracting the outputs, and $f(4) = 7$ by the pattern.

C. $f^{-1}(7) = 2$ because $f(2) = 7$, and then $f(2) = 7$, so $f(f^{-1}(7)) = 7$.

D. $f^{-1}(7) = 3$ because $f(1) = 3$, and then $f(3) = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

2898

QUESTION ID
S77-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. $f^{-1}(7) = 2$ because $f(2) = 7$, and then $f(2) = 7$, so $f(f^{-1}(7)) = 7$.

GRADING NOTE Read the table backward, then forward.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c • ADVANCED EXTENSION (+)

Read values of an inverse function from graphs and tables

MAIN	Algebra II main question	**HSF-BF.B.4.c**
BACKWARD 1	Inverse functions swap inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 2	Read points from graphs and tables	**8.F.A.1**
BACKWARD 3	One-to-one functions	**HSF-BF.B.4.b**
FORWARD 1	Estimate inverse values between data points and sketch inverse...	**HSF-BF.B.4.c**
FORWARD 2	Restrict the domain to make a function invertible	**HSF-BF.B.4.d**

SET 78 • HSF-BF.B.4.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

The table shows f. Find $f^{-1}(15)$.

x	0	1	2	3	4
f(x)	1	3	7	15	31

$f^{-1}(15) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

2901

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE $f(3) = 15$.

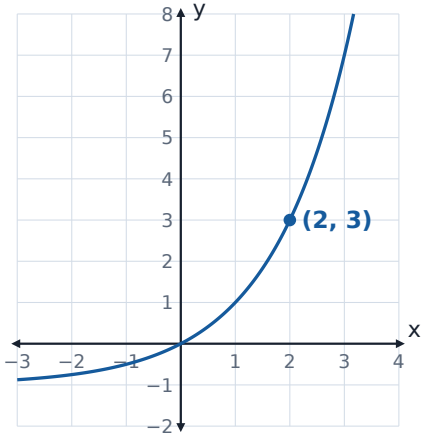
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = 2^x - 1$ is shown. Use it to find $f^{-1}(3)$.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

2902

QUESTION ID
S78-M-Q2

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. The point (2, 3) is on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a decreasing function g. Find $g^{-1}(10)$ and $g^{-1}(g(2))$.

x	1	2	3	4	5
g(x)	40	25	16	10	4

$g^{-1}(10) =$ _____

$g^{-1}(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

2903

QUESTION ID

S78-M-Q3

TYPE

Short answer

CORRECT ANSWER

4; 2

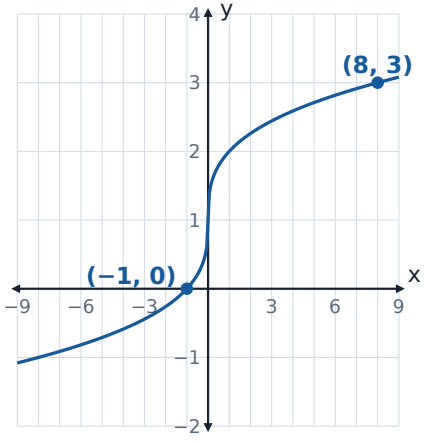
GRADING NOTE Form required: both. g(4) = 10; g^{-1} undoes g.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = \sqrt[3]{x} + 1$ is shown. Use it to find $f^{-1}(0)$ and $f^{-1}(3)$.



$f^{-1}(0) =$ _____

$f^{-1}(3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

2904

QUESTION ID
S78-M-Q4

TYPE
Short answer

CORRECT ANSWER
−1; 8

GRADING NOTE Form required: both. The points (−1, 0) and (8, 3) are on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The table gives a plane's altitude A (m) t minutes after takeoff. Find $A^{-1}(1200)$ and interpret it.

t (min)	0	1	2	3	4	5
A (m)	0	250	550	880	1200	1500

A. $A^{-1}(1200) = 4$: the plane climbs 1200 m every minute.

B. $A^{-1}(1200) = 1500$: the altitude one minute after reaching 1200 m.

C. $A^{-1}(1200) = 4$: the plane reaches 1200 m four minutes after takeoff.

D. $A^{-1}(1200) = 1/1200$: the reciprocal of the altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

2905

QUESTION ID

S78-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $A^{-1}(1200) = 4$: the plane reaches 1200 m four minutes after takeoff.

GRADING NOTE $A(4) = 1200$; the inverse gives the time for an altitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 1

Inverse functions swap inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

If $f(3) = 15$, what is $f^{-1}(15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

2907

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x - 1$.

A. $f^{-1}(x) = \log_2 x + 1$

B. $f^{-1}(x) = 2^{-x} + 1$

C. $f^{-1}(x) = \log_2(x - 1)$

D. $f^{-1}(x) = \log_2(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

2908

QUESTION ID

S78-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D.

$f^{-1}(x) = \log_2(x + 1)$

GRADING NOTE $x + 1 = 2^y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f^{-1}(y)$ is the input of f that gives the output y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

2909

QUESTION ID

S78-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If (5, 4) is on the graph of g, which point is on g^{-1}?

A. (5, 4)

B. (1/5, 1/4)

C. (−5, −4)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

2910

QUESTION ID

S78-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 5)

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the formula to find $f^{-1}(3)$ for $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

2911

QUESTION ID

S78-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 2

Read points from graphs and tables • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. What is f(2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

2913

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane's altitude A (m) at t = 0, 1, 2, 3, 4 minutes is 0, 250, 550, 880, 1200. What is A(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

2914

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

880 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

True or false?

A function gives exactly one output for each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

2915

QUESTION ID

S78-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Which input of g gives 16?

A. 16

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

2916

QUESTION ID

S78-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE Read backward.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph passes through (8, 3). What is the output at input 8?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

2917

QUESTION ID

S78-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 3

One-to-one functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Why does a strictly increasing function have an inverse function?

A. Its graph passes through the origin, so it can be reflected over y = x.

B. Different inputs give different outputs, so each output comes from exactly one input.

C. Its outputs are always positive, so each output can be undone.

D. Its domain is all real numbers, so every input has an output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

2919

QUESTION ID

S78-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Different inputs give different outputs, so each output comes from exactly one input.

GRADING NOTE Strictly increasing implies one-to-one. The others name properties (passing through the origin, positive outputs, domain) that do not make a function one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

Does a function with f(1) = 4 and f(3) = 4 have an inverse function? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

2920

QUESTION ID

S78-B3-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE The output 4 has two inputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-B3-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

True or false?

A decreasing function can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

2921

QUESTION ID

S78-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Still one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

Which table has an inverse function?

A. x: 1, 2, 3; y: 5, 7, 5

B. x: 1, 2, 3; y: 5, 7, 9

C. x: 1, 2, 3; y: 0, 1, 0

D. x: 1, 2, 3; y: 2, 2, 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

2922

QUESTION ID

S78-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 2, 3; y: 5, 7, 9

GRADING NOTE Distinct outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(2) = 7 and f(3) = 15. Use the table to find $f^{-1}(15)$, then $f(f^{-1}(15))$.

$f^{-1}(15) =$ _____

$f(f^{-1}(15)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

2923

QUESTION ID

S78-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3; 15

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 1

Estimate inverse values between data points and sketch inverse graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A plane's altitude is 880 m at t = 3 min and 1200 m at t = 4 min. Assume it climbs at a constant rate between these times. Estimate the time when it reaches 1000 m, to the nearest tenth of a minute.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

2925

QUESTION ID
S78-F1-Q1

TYPE
Short answer

CORRECT ANSWER
3.4

GRADING NOTE 3 + 120/320 = 3.375, which rounds to 3.4.

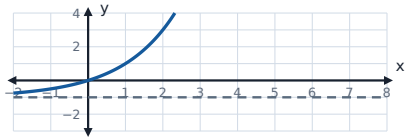
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

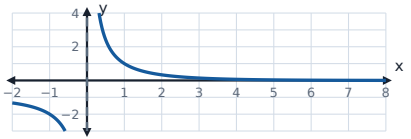
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows the inverse of $f(x) = 2^x - 1$?

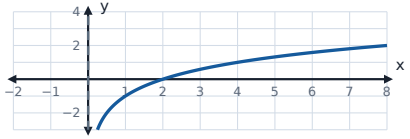
A



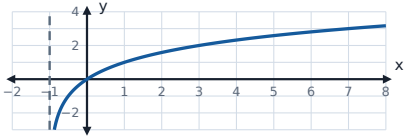
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

2926

QUESTION ID

S78-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Swap the coordinates of points of f: (0, 0), (1, 1), (2, 3) on f give (0, 0), (1, 1), (3, 2) on f^{-1}; the asymptote $y = -1$ becomes $x = -1$; $f^{-1}(x) = \log_2(x + 1)$. The others show f itself, the reciprocal $1/f(x)$, and $\log_2 x - 1$ (operations undone in the wrong order).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

True or false?

The graph of f^{-1} is the reflection of the graph of f across y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

2927

QUESTION ID
S78-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Swap coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A decreasing function has $g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Estimate $g^{-1}(20)$.

A. between 1 and 2

B. between 2 and 3

C. between 4 and 5

D. between 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

2928

QUESTION ID

S78-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. between 2 and 3

GRADING NOTE $g(2) = 25$, $g(3) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $(f^{-1})^{-1}(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

2929

QUESTION ID
S78-F1-Q5

TYPE
Short answer

CORRECT ANSWER
f(2) = 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 2

Restrict the domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 3)^2$ is not invertible on all real numbers. Which restriction and inverse are correct?

A. $x \geq 3$; $f^{-1}(x) = 3 + \sqrt{x}$

B. $x \geq 3$; $f^{-1}(x) = 3 - \sqrt{x}$

C. $x \geq 0$; $f^{-1}(x) = \sqrt{x} + 3$

D. $x \leq 3$; $f^{-1}(x) = 3 + \sqrt{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

2931

QUESTION ID

S78-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $x \geq 3$; $f^{-1}(x) = 3 + \sqrt{x}$

GRADING NOTE On $x \geq 3$, $y = (x - 3)^2$ gives $x = 3 + \sqrt{y}$. On $x \leq 3$ the inverse is $3 - \sqrt{x}$; $x \geq 0$ is not one-to-one ($f(2) = f(4)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(x) = x^2$ has $h(-2) = 4$, $h(0) = 0$, and $h(2) = 4$. Why is $h^{-1}(4)$ not defined, and how can the domain be restricted to fix it?

A. 4 is the output of two inputs (−2 and 2); restrict h to $x \geq 0$ (or to $x \leq 0$).

B. 0 has no inverse, so remove $x = 0$ from the domain of h.

C. 4 is the output of two inputs (−2 and 2); restrict h to $-2 \leq x \leq 2$.

D. $h^{-1}(4) = 1/4$, so it is defined, and no restriction is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

2932

QUESTION ID
S78-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. 4 is the output of two inputs (−2 and 2); restrict h to $x \geq 0$ (or to $x \leq 0$).

GRADING NOTE Each half of the parabola is one-to-one; $-2 \leq x \leq 2$ still contains both inputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

True or false?

Restricting the domain can turn a non-invertible function into an invertible one.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

2933

QUESTION ID
S78-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE One-to-one on the piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ restricted to $x \le 0$ has inverse

A. $\sqrt{x}$

B. x^2

C. $-\sqrt{x}$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

2934

QUESTION ID

S78-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-\sqrt{x}$

GRADING NOTE Negative outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 3)^2$ has range $y \geq 0$. When its domain is restricted to make it invertible, why should the restricted function still take every output value $y \geq 0$?

A. So the restricted function has more outputs than the original function, which makes the inverse larger.

B. So the inverse is defined for every output of the original function; for example, $x \geq 3$ still gives every $y \geq 0$.

C. Because the inverse function must be identical to the original function on the restricted domain.

D. Because a restricted function must always have exactly the same domain as the original function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

2935

QUESTION ID
S78-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. So the inverse is defined for every output of the original function; for example, x ≥ 3 still gives every y ≥ 0.

GRADING NOTE Keeping the original range keeps the inverse defined on all of it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d • ADVANCED EXTENSION (+)

Restrict a domain to make a function invertible and write the inverse

MAIN	Algebra II main question	**HSF-BF.B.4.d**
BACKWARD 1	Find the vertex of a quadratic	**HSF-IF.C.8.a**
BACKWARD 2	Write inverses of one-to-one functions	**HSF-BF.B.4.a**
BACKWARD 3	Domains and ranges	**HSF-IF.B.5**
FORWARD 1	Compare different restrictions	**HSF-BF.B.4.d**
FORWARD 2	Restrict trigonometric functions to define inverses	**HSF-TF.B.6**

SET 79 • HSF-BF.B.4.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x - 3)^2$ to $x \geq 3$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 3$

B. $f^{-1}(x) = \sqrt{(x + 3)},\ x \geq -3$

C. $f^{-1}(x) = \sqrt{x} - 3,\ x \geq 0$

D. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

2938

QUESTION ID

S79-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. $f^{-1}(x) = \sqrt{x} + 3,\ x \geq 0$

GRADING NOTE $y = (x - 3)^2$ with $x \geq 3$ gives $x = \sqrt{y} + 3$; the domain of f^{-1} is the range of f, $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = x^2 + 1$ to $x \le 0$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = \sqrt{(x - 1)}, x \ge 1$

B. $f^{-1}(x) = -\sqrt{(x + 1)}, x \ge -1$

C. $f^{-1}(x) = -\sqrt{(x - 1)}, x \ge 1$

D. $f^{-1}(x) = -\sqrt{(x - 1)}, x \le 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

2939

QUESTION ID

S79-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = -\sqrt{(x - 1)}, x \ge 1$

GRADING NOTE The outputs of f^{-1} must be ≤ 0, so take the negative root; the domain is the range of f, $x \ge 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = |x + 2|$ to $x \geq -2$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = x + 2,\ x \geq -2$

B. $f^{-1}(x) = x - 2,\ x \geq 0$

C. $f^{-1}(x) = -x - 2,\ x \geq 0$

D. $f^{-1}(x) = x - 2,\ x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

2940

QUESTION ID

S79-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f^{-1}(x) = x - 2,\ x \geq 0$

GRADING NOTE On $x \geq -2$, $f(x) = x + 2$ with range $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x + 1)^2 - 4$ to $x \geq -1$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -4$

B. $f^{-1}(x) = -1 - \sqrt{(x + 4)}, x \geq -4$

C. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -1$

D. $f^{-1}(x) = 1 + \sqrt{(x - 4)}, x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

2941

QUESTION ID

S79-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $f^{-1}(x) = -1 + \sqrt{(x + 4)}, x \geq -4$

GRADING NOTE $x + 1 = \sqrt{(y + 4)}$ for $x \geq -1$; the range of f is $y \geq -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = x^4$ to $x \geq 0$. Which is f^{-1} with its domain?

A. $f^{-1}(x) = x/4,\ x \geq 0$

B. $f^{-1}(x) = \pm\sqrt[4]{x},\ x \geq 0$

C. $f^{-1}(x) = \sqrt[4]{x},\ x \geq 0$

D. $f^{-1}(x) = 1/x^4,\ x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

2942

QUESTION ID

S79-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = \sqrt[4]{x},\ x \geq 0$

GRADING NOTE The nonnegative fourth root; ± is not a function; $1/x^4$ is a reciprocal, not an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 1

Find the vertex of a quadratic • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertex of $y = (x + 1)^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

2944

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, −4)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertex of $y = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

2945

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

True or false?

A parabola is increasing on one side of its vertex and decreasing on the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

2946

QUESTION ID

S79-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The vertex of $y = (x - 3)^2$ is

A. (0, 9)

B. (0, 3)

C. (−3, 0)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

2947

QUESTION ID

S79-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (3, 0)

GRADING NOTE h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

On which side of the vertex is $y = (x - 3)^2$ increasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

2948

QUESTION ID

S79-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x ≥ 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 2

Write inverses of one-to-one functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = (x - 3)^2$ for x when $x \geq 3$.

A. $x = 3 - \sqrt{y}$

B. $x = \sqrt{(y + 3)}$

C. $x = \sqrt{y} - 3$

D. $x = 3 + \sqrt{y}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

2950

QUESTION ID

S79-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. $x = 3 + \sqrt{y}$

GRADING NOTE $x - 3 = \sqrt{y}$ because $x - 3 \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = x^2 + 1$ for x when $x \leq 0$.

A. $x = \sqrt{(y - 1)}$

B. $x = -\sqrt{y} - 1$

C. $x = -\sqrt{(y + 1)}$

D. $x = -\sqrt{(y - 1)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

2951

QUESTION ID

S79-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $x = -\sqrt{(y - 1)}$

GRADING NOTE $x^2 = y - 1$ and $x \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When $x \leq 0$, the solution of $x^2 = 4$ is $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

2952

QUESTION ID

S79-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Choose the negative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving $y = x + 2$ $(x \geq -2)$ for x gives

A. $2 - y$

B. $y + 2$

C. $y - 2$

D. $-y - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

2953

QUESTION ID
S79-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. y − 2

GRADING NOTE Linear piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = x^4$ for x when $x \geq 0$. Give x as a power of y, using ^ for exponents.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

2954

QUESTION ID

S79-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$y^{1/4}$

GRADING NOTE The nonnegative fourth root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 3

Domains and ranges • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = (x - 3)^2$ for $x \geq 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

2956

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = (x + 1)^2 - 4$ for $x \geq -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

2957

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$y \geq -4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

True or false?

The domain of the inverse is the range of the restricted function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

2958

QUESTION ID
S79-B3-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The range of $x^2 + 1$ for $x \leq 0$ is

A. $y \geq 0$

B. $y \geq 1$

C. all reals

D. $y \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

2959

QUESTION ID

S79-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y \geq 1$

GRADING NOTE Minimum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $|x + 2|$ for $x \geq -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

2960

QUESTION ID

S79-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$y \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

2960

SET 79 • HSF-BF.B.4.d

FORWARD 1

Compare different restrictions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x - 3)^2$ to $x \leq 3$. Which is the inverse with its domain?

A. $f^{-1}(x) = 3 - \sqrt{x}$, $x \leq 3$

B. $f^{-1}(x) = 3 + \sqrt{x}$, $x \geq 0$

C. $f^{-1}(x) = 3 - \sqrt{x}$, $x \geq 0$

D. $f^{-1}(x) = \sqrt{x} - 3$, $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

2962

QUESTION ID

S79-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = 3 - \sqrt{x}$, $x \geq 0$

GRADING NOTE For $x \leq 3$, $x - 3 = -\sqrt{y}$. The inverse's domain is the restricted function's range, $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -5(t - 2)^2 + 20$ for $0 \le t \le 4$. Which restriction and inverse give the time as a function of height on the way down?

A. $2 \le t \le 4$; $t = 2 + \sqrt{((20 - h)/5)}$

B. $2 \le t \le 4$; $t = 2 - \sqrt{((20 - h)/5)}$

C. $0 \le t \le 4$; $t = 2 + \sqrt{((20 - h)/5)}$

D. $0 \le t \le 2$; $t = 2 - \sqrt{((20 - h)/5)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

2963

QUESTION ID

S79-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $2 \le t \le 4$; $t = 2 + \sqrt{((20 - h)/5)}$

GRADING NOTE The ball falls after the vertex at $t = 2$, where $t - 2 \ge 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

True or false?

A function can have more than one valid restriction that makes it invertible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

2964

QUESTION ID

S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Either side of the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which restriction does not make $f(x) = x^2$ invertible?

A. $x \geq 0$

B. $x \leq 0$

C. $-1 \leq x \leq 1$

D. $1 \leq x \leq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

2965

QUESTION ID

S79-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-1 \leq x \leq 1$

GRADING NOTE Contains both ±1/2, for example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is the restriction $x \geq 0$ natural for $f(x) = x^2$ in the area formula $A = s^2$?

A. x^2 is always increasing.

B. Side lengths are nonnegative.

C. Squares have four sides.

D. Areas can be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

2966

QUESTION ID

S79-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Side lengths are nonnegative.

GRADING NOTE Context restricts the domain. The others are false or irrelevant: x^2 is not always increasing, the number of sides does not restrict s, and areas are not negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 2

Restrict trigonometric functions to define inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Why is sin x not invertible on all real numbers?

A. It repeats values (for example sin 0 = sin π = 0), so it is not one-to-one.

B. Its outputs are larger than 1 for large x, outside the range of an inverse.

C. It is not defined for negative x, so an inverse would be incomplete.

D. It is an odd function, and odd functions never have an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

2968

QUESTION ID

S79-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. It repeats values (for example sin 0 = sin π = 0), so it is not one-to-one.

GRADING NOTE A function must be one-to-one to have an inverse. The others are false: sin x stays between −1 and 1, is defined for negative x, and being odd does not prevent an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

What is the standard restricted domain of sin x, and why does it work?

A. $0 \le x \le \pi$: sin is increasing there and takes every value from -1 to 1 once.

B. $-\pi/2 \le x \le \pi/2$: sin is decreasing there and takes every value from 0 to 1 once.

C. $0 \le x \le 2\pi$: sin takes every value from -1 to 1 there, which is all that is needed.

D. $-\pi/2 \le x \le \pi/2$: sin is increasing there and takes every value from -1 to 1 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

2969

QUESTION ID

S79-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $-\pi/2 \le x \le \pi/2$: sin is increasing there and takes every value from -1 to 1 once.

GRADING NOTE On $[-\pi/2, \pi/2]$ sin rises from -1 to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

True or false?

cos x is restricted to $0 \leq x \leq \pi$ to define arccos.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

2970

QUESTION ID
S79-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Decreasing there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arcsin(1) equals

A. 0

B. 3π/2

C. π/2

D. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

2971

QUESTION ID

S79-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π/2

GRADING NOTE In the restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

Why would $0 \le x \le \pi$ not work as a restricted domain for an inverse of sine?

A. sin is not defined at $x = \pi$, so the interval is not closed.

B. sin is negative everywhere on that interval, so it misses positive values.

C. sin increases then decreases there, so values repeat ($\sin(\pi/6) = \sin(5\pi/6)$).

D. sin is increasing on that whole interval, so it is one-to-one there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

2972

QUESTION ID

S79-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. sin increases then decreases there, so values repeat ($\sin(\pi/6) = \sin(5\pi/6)$).

GRADING NOTE The function must be one-to-one on the restricted domain. The others are false: sin is defined at π, is nonnegative on [0, π], and is not increasing on the whole interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSF-BF.B.5 • ADVANCED EXTENSION (+)

Use the inverse relationship of exponents and logarithms to solve equations

MAIN	Algebra II main question	**HSF-BF.B.5**
BACKWARD 1	Write exact logarithmic solutions of exponential equations	**HSF-LE.A.4**
BACKWARD 2	Check for extraneous solutions	**HSA-REI.A.2**
BACKWARD 3	Domains of logarithmic expressions	**HSF-IF.B.5**
FORWARD 1	Use log properties and change of base in equations	**HSF-BF.B.5**
FORWARD 2	Check logarithmic solutions graphically	**HSA-REI.D.11**

SET 80 • HSF-BF.B.5

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_3 x = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

2975

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

81

GRADING NOTE Any equivalent form. $x = 3^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\log_5(x - 1) = 4$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

2976

QUESTION ID

S80-M-Q2

TYPE

Short answer

CORRECT ANSWER

26

GRADING NOTE Any equivalent form. $\log_5(x - 1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ln(x + 2) = 1 exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

2977

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

e − 2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2 x + \log_2(x - 2) = 3$. The equation $x(x - 2) = 8$ gives two candidates. Give the actual solution and the extraneous value.

Solution: x = _____

Extraneous value: x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

2978

QUESTION ID

S80-M-Q4

TYPE

Short answer

CORRECT ANSWER

4; −2

GRADING NOTE Form required: both. $x^2 - 2x - 8 = 0$ gives 4 or −2; $\log_2(-2)$ is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x − log(x − 9) = 1 (base 10) and check.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

2979

QUESTION ID
S80-M-Q5

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Form required: the solution with a check. x/(x − 9) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 1

Write exact logarithmic solutions of exponential equations • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3^x = 81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

2981

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^x = 5$ exactly.

A. $x = 5/e$

B. $x = \ln 5$

C. $x = e^5$

D. $x = \log 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

2982

QUESTION ID

S80-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $x = \ln 5$

GRADING NOTE ln is the inverse of e^x; log 5 is base 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

2982

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log_b y = x means $b^x = y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

2983

QUESTION ID

S80-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$5^2 = 25$ written as a logarithm is

A. $\log_5 2 = 25$

B. $\log_2 25 = 5$

C. $\log_{25} 5 = 2$

D. $\log_5 25 = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

2984

QUESTION ID

S80-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $\log_5 25 = 2$

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $10^x = 1000$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

2985

QUESTION ID

S80-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 2

Check for extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = -2$ in $\log_2 x + \log_2(x - 2) = 3$.

A. $\log_2(-2) = 1$, so the left side is 3 and $x = -2$ is a solution.

B. $\log_2(-2) = -1$, so $x = -2$ is a solution.

C. $\log_2(-2)$ is undefined, so $x = -2$ is not a solution.

D. $(-2)(-4) = 8 = 2^3$, so $x = -2$ is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

2987

QUESTION ID

S80-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. $\log_2(-2)$ is undefined, so $x = -2$ is not a solution.

GRADING NOTE Logarithms of negative numbers are undefined; the product check skips the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 4$ in $\log_2 x + \log_2(x - 2) = 3$: find the value of the left side, then say whether $x = 4$ is a solution.

Left side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

2988

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3; Yes

GRADING NOTE Form required: both. $\log_2 4 + \log_2 2 = 2 + 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

True or false?

Combining logs can introduce solutions outside the original domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

2989

QUESTION ID

S80-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE log a + log b = log(ab) needs a, b > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is excluded from log(x − 9)?

A. x = 20

B. x = 100

C. x = 5

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

2990

QUESTION ID

S80-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

GRADING NOTE x − 9 > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

2990

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x(x - 2) = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

2991

QUESTION ID

S80-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4 or −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 3

Domains of logarithmic expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log_5(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

2993

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x > 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of ln(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

2994

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x > −2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log 0 = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

2995

QUESTION ID

S80-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No power of 10 equals 0, so log 0 is undefined (log 1 = 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log_2 x + \log_2(x - 2)$ is

A. $x > 2$

B. $x \neq 2$

C. $x > 0$

D. all reals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

2996

QUESTION ID

S80-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x > 2

GRADING NOTE Both inputs positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log x - \log(x - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

2997

QUESTION ID

S80-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x > 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 1

Use log properties and change of base in equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2(x + 6) - \log_2 x = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

2999

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $(x + 6)/x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_4 x = \log_2 3$. (Hint: change of base.)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

3000

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $\log_4 x = (\log_2 x)/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_2(x + 6) - \log_2 x = \log_2((x + 6)/x)$ for $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

3001

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Quotient property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\log(x + 1) + \log(x - 1) = \log 8$ has solution

A. no solution

B. $x = 8$

C. $x = 3$

D. $x = -3$ and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

3002

QUESTION ID

S80-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x = 3$

GRADING NOTE $x^2 - 1 = 8$; -3 rejected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

Why does $\log_2(x - 5) = \log_2(1 - x)$ have no solution?

A. Both sides are defined only if $x > 5$ and $x < 1$ at the same time, which is impossible.

B. Setting $x - 5 = 1 - x$ gives $x = 3$, which is a solution, so the equation does have one.

C. The equation has a solution only when x is negative.

D. Logarithms with base 2 can never be equal to each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

3003

QUESTION ID

S80-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Both sides are defined only if x > 5 and x < 1 at the same time, which is impossible.

GRADING NOTE The candidate x = 3 makes both arguments negative. The others accept x = 3 without checking the domain, invent a condition about negative x, or claim logarithms cannot be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 2

Check logarithmic solutions graphically • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

How do the graphs of $y = \log_2 x + \log_2(x - 2)$ and $y = 3$ show that the only solution is $x = 4$?

A. They intersect only at $x = 4$; the left graph exists only for $x > 2$, so there is no point at $x = -2$.

B. They intersect at $x = 4$ and at $x = -2$, so both values are solutions of the equation.

C. They never intersect, so $x = 4$ must come from the algebra only.

D. They intersect at $y = 4$, which is the solution of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

3005

QUESTION ID
S80-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. They intersect only at x = 4; the left graph exists only for x > 2, so there is no point at x = −2.

GRADING NOTE Graphs show only points in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to approximate the solution of $\ln x = 2 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

3006

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.56

GRADING NOTE Accept 1.55 to 1.56 (root 1.5571).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of a log equation corresponds to no intersection point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

3007

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outside the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of y = log x and y = 1 meet at

A. x = e

B. x = 10

C. x = 1

D. x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

3008

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = 10

GRADING NOTE log 10 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can ln x = 2 − x not be solved exactly with logarithm rules?

A. ln x is undefined for every x, so the equation has no solution at all.

B. Logarithm rules apply only to equations with base-10 logarithms.

C. The equation has infinitely many solutions, so no single exact value exists.

D. x appears both inside a logarithm and outside it, so no algebraic step isolates x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

3009

QUESTION ID

S80-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. x appears both inside a logarithm and outside it, so no algebraic step isolates x.

GRADING NOTE Such equations are solved with graphs or tables. The others are false: ln x is defined for x > 0, the equation has one solution, and the rules apply to every base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSF-TF.A.3 • ADVANCED EXTENSION (+)

Derive values from special triangles and relate π − x, π + x, 2π − x to x

MAIN	Algebra II main question	**HSF-TF.A.3**
BACKWARD 1	Special right triangle side ratios	**HSG-SRT.C.6**
BACKWARD 2	Unit-circle values in all quadrants	**HSF-TF.A.2**
BACKWARD 3	The Pythagorean Theorem	**8.G.B.7**
FORWARD 1	Derive more special values and the reflection identities	**HSF-TF.A.3**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 81 • HSF-TF.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A 30°−60°−90° triangle with hypotenuse 1 is half of an equilateral triangle with side 1. Which derivation of sin(π/3) and cos(π/3) is correct?

A. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the long leg, so sin(π/3) = √3/2 and cos(π/3) = 1/2.

B. The long leg is √(1 + 1/4) = √5/2, so sin(π/3) = √5/2.

C. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the short leg, so sin(π/3) = 1/2 and cos(π/3) = √3/2.

D. The legs are 1/2 and 1 − 1/2 = 1/2, so sin(π/3) = cos(π/3) = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

3012

QUESTION ID

S81-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the long leg, so sin(π/3) = √3/2 and cos(π/3) = 1/2.

GRADING NOTE Halving the base gives 1/2; the Pythagorean Theorem gives √3/2; the larger angle is opposite the longer leg. The others put π/3 opposite the short leg, subtract lengths instead of using the Pythagorean Theorem, or add instead of subtracting squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which derivation of tan(π/4) from a 45°−45°−90° triangle is correct?

A. The hypotenuse is twice a leg, so tan(π/4) = 1/2.

B. tan(π/4) = sin(π/4) + cos(π/4) = √2.

C. The legs are 1 and √2, so tan(π/4) = opposite/adjacent = 1/√2 = √2/2.

D. The two legs are equal, so tan(π/4) = opposite/adjacent = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

3013

QUESTION ID

S81-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. The two legs are equal, so tan(π/4) = opposite/adjacent = 1.

GRADING NOTE Equal legs make the ratio 1. The others use the hypotenuse as a leg, add instead of dividing, or use a 30°−60°−90° fact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

Which uses sin(π − x) = sin x to find sin(5π/6) exactly, with a correct unit-circle explanation?

A. sin(5π/6) = −sin(π/6) = −1/2: the points for x and π − x are reflections over the x-axis, so their y-coordinates are opposite.

B. sin(5π/6) = sin(π/6) = 1/2: the points for x and π − x are reflections over the y-axis, so they have the same y-coordinate.

C. sin(5π/6) = cos(π/6) = √3/2: the point for π − x is a quarter turn from the point for x, so y becomes x.

D. sin(5π/6) = sin(π/6) = 1/2: the points for x and π − x are reflections over the x-axis, so they have the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

3014

QUESTION ID

S81-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. sin(5π/6) = sin(π/6) = 1/2: the points for x and π − x are reflections over the y-axis, so they have the same y-coordinate.

GRADING NOTE (cos x, sin x) and (−cos x, sin x) mirror over the y-axis. The others name the x-axis reflection (which changes the sign of y), or treat π − x as a quarter turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use cos(π + x) = −cos x to find cos(7π/6) exactly.

cos(7π/6) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

3015

QUESTION ID

S81-M-Q4

TYPE

Short answer

CORRECT ANSWER

−√3/2

GRADING NOTE cos(7π/6) = −cos(π/6) = −√3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\tan(2\pi - x) = -\tan x$ to find $\tan(5\pi/3)$ exactly.

$\tan(5\pi/3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

3016

QUESTION ID

S81-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-\sqrt{3}$

GRADING NOTE $\tan(5\pi/3) = -\tan(\pi/3) = -\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 1

Special right triangle side ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A 30°−60°−90° triangle has hypotenuse 2. Give the lengths of its legs exactly.

Short leg: _____

Long leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

3018

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1; √3

GRADING NOTE Form required: both. Half of an equilateral triangle with side 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the hypotenuse of a 45°−45°−90° triangle with legs 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

3019

QUESTION ID
S81-B1-Q2

TYPE
Short answer

CORRECT ANSWER
√2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

True or false?

In a 30°−60°−90° triangle, the shortest side is half the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

3020

QUESTION ID

S81-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Half of an equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos 30° equals

A. √3

B. √2/2

C. 1/2

D. √3/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

3021

QUESTION ID

S81-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √3/2

GRADING NOTE Adjacent/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give tan 60°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

3022

QUESTION ID

S81-B1-Q5

TYPE

Short answer

CORRECT ANSWER

√3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 2

Unit-circle values in all quadrants • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

3024

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−√3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(7π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

3025

QUESTION ID

S81-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

True or false?

sin is positive in quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

3026

QUESTION ID

S81-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $y > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quadrant contains 5π/3?

A. II

B. I

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

3027

QUESTION ID

S81-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. IV

GRADING NOTE Between 3π/2 and 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give tan(π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

3028

QUESTION ID

S81-B2-Q5

TYPE

Short answer

CORRECT ANSWER

√3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 3

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 1 and leg 1/2. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

3030

QUESTION ID
S81-B3-Q1

TYPE
Short answer

CORRECT ANSWER
√3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right isosceles triangle has hypotenuse 1. Find each leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

3031

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(1/2)^2 + (\sqrt{3}/2)^2 = 2.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

3032

QUESTION ID

S81-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1/4 + 3/4 = 1.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

Legs 1 and 1 give hypotenuse

A. 2

B. √3

C. √2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

3033

QUESTION ID

S81-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. √2

GRADING NOTE √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find x if $x^2 + x^2 = 1$ ($x > 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

3034

QUESTION ID

S81-B3-Q5

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 1

Derive more special values and the reflection identities • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which derivation of tan(π/6) from a special triangle is correct?

A. The opposite side is half the hypotenuse, so tan(π/6) = 1/2.

B. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg 1/2 and adjacent to the leg √3/2, so tan(π/6) = (1/2)/(√3/2) = 1/√3 = √3/3.

C. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg √3/2 and adjacent to the leg 1/2, so tan(π/6) = (√3/2)/(1/2) = √3 exactly.

D. tan(π/6) = (1/2)/(√3/2) = √3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

3036

QUESTION ID

S81-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg 1/2 and adjacent to the leg √3/2, so tan(π/6) = (1/2)/(√3/2) = 1/√3 = √3/3.

GRADING NOTE Opposite over adjacent, then simplify (1/2)/(√3/2) = 1/√3 = √3/3. The others use the angle π/3, give sin(π/6) instead, or simplify the quotient incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle argument explains why $\cos(\pi - x) = -\cos x$?

A. The points for x and $\pi - x$ are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

B. $\cos(\pi - \pi/3) = -1/2 = -\cos(\pi/3)$, so the rule always holds.

C. $\cos(\pi - x) = \cos \pi - \cos x = -1 - \cos x$.

D. The points for x and $\pi - x$ are reflections of each other over the x-axis, so their x-coordinates (the cosines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

3037

QUESTION ID

S81-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The points for x and π − x are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

GRADING NOTE Reflection over the y-axis changes the sign of x and keeps y. The others name the wrong reflection (it keeps x), distribute cosine over a difference, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(\pi + x) = -\sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

3038

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotation by π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos(2π − x) equals

A. cos x

B. −cos x

C. sin x

D. −sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

3039

QUESTION ID

S81-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos x

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\sin(2\pi - x) = -\sin x$ to find $\sin(11\pi/6)$ exactly.

$\sin(11\pi/6) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

3040

QUESTION ID

S81-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$-1/2$

GRADING NOTE $\sin(11\pi/6) = -\sin(\pi/6) = -1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle reasoning explains why $\sin(-x) = -\sin x$?

A. The points for x and −x are opposite points, so their x-coordinates are equal.

B. The points for x and −x are reflections over the y-axis, so their y-coordinates are opposites.

C. The points for x and −x are reflections over the x-axis, so their y-coordinates are opposites.

D. The points for x and −x are the same point, so their y-coordinates are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

3042

QUESTION ID

S81-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The points for x and −x are reflections over the x-axis, so their y-coordinates are opposites.

GRADING NOTE Equal arcs in opposite directions from (1, 0) mirror over the x-axis. The others call the points opposite or the same, or name the y-axis reflection, which keeps the y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

Is cos an even or odd function, and why?

A. Even: the points for x and −x have the same y-coordinate, so cos(−x) = cos x.

B. Even: the points for x and −x have the same x-coordinate, so cos(−x) = cos x.

C. Odd: the points for x and −x have opposite x-coordinates, so cos(−x) = −cos x.

D. Neither: cos(−x) is not related to cos x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

3043

GRADING NOTE Reflection over the x-axis keeps the x-coordinate.

QUESTION ID
S81-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. Even: the points for x and −x have the same x-coordinate, so cos(−x) = cos x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

True or false?

$\sin(x + 2\pi) = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

3044

QUESTION ID

S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

tan(−π/4) equals

A. undefined

B. −1

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

3045

QUESTION ID

S81-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1

GRADING NOTE Odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using $2\pi - x = -x + 2\pi$, which reasoning explains why $\tan(2\pi - x) = -\tan x$?

A. $\tan(-x + 2\pi) = \tan(-x)$ by periodicity, and tan is odd, so $\tan(-x) = -\tan x$.

B. $\tan(-x + 2\pi) = \tan x$ by periodicity, and tan is even.

C. $\tan(-x + 2\pi) = \tan(-x) + \tan(2\pi) = -\tan x + 1$.

D. $\tan(2\pi - x) = 2\pi - \tan x$, because tan distributes over subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

3046

QUESTION ID

S81-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $\tan(-x + 2\pi) = \tan(-x)$ by periodicity, and tan is odd, so $\tan(-x) = -\tan x$.

GRADING NOTE Period π (so also 2π) and oddness.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSF-TF.A.4 • ADVANCED EXTENSION (+)

Explain the symmetry and periodicity of trigonometric functions with the unit circle

MAIN	Algebra II main question	**HSF-TF.A.4**
BACKWARD 1	Explain the unit-circle definition	**HSF-TF.A.2**
BACKWARD 2	Even and odd functions	**HSF-BF.B.3**
BACKWARD 3	Reflect points over the axes	**8.G.A.3**
FORWARD 1	Use periodicity and symmetry to evaluate and simplify	**HSF-TF.A.4**
FORWARD 2	Restrict trigonometric functions so they have inverses	**HSF-TF.B.6**

SET 82 • HSF-TF.A.4

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why $\sin(-x) = -\sin x$?

A. The points for x and −x are (a, b) and (−a, b), reflections of each other over the y-axis, so their sines are opposites.

B. $\sin(-\pi/6) = -1/2 = -\sin(\pi/6)$, so the rule always holds.

C. The points for x and −x are (a, b) and (a, −b), reflections over the x-axis, so the sines b and −b are opposites.

D. A negative sign can always be moved outside a function, so $\sin(-x) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

3049

QUESTION ID

S82-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The points for x and −x are (a, b) and (a, −b), reflections over the x-axis, so the sines b and −b are opposites.

GRADING NOTE Equal arcs in opposite directions from (1, 0) give points that mirror over the x-axis. The others name the wrong reflection, use a false general rule ($\cos(-x) = \cos x$, not $-\cos x$), or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why $\cos(x + 2\pi) = \cos x$?

A. Adding 2π moves the point to the opposite side of the unit circle, and opposite points on the circle have the same x-coordinate, so the cosine is the same.

B. $\cos(x + 2\pi) = \cos x + \cos 2\pi = \cos x + 1$.

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

D. $\cos(0 + 2\pi) = 1 = \cos 0$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

3050

QUESTION ID
S82-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

GRADING NOTE The circumference is 2π, so x and $x + 2\pi$ give the same point. The others distribute cosine over a sum, describe the wrong motion, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why $\tan(x + \pi) = \tan x$?

A. Adding π gives the point (a, −b), a reflection over the x-axis, so $\tan(x + \pi) = -b/a = -\tan x$, which is not tan x.

B. Adding π gives the opposite point (−a, −b) of (a, b), and $\tan = (-b)/(-a) = b/a$, the same ratio.

C. Adding π travels a full turn, so the point is the same.

D. $\tan(\pi/4 + \pi) = 1 = \tan(\pi/4)$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

3051

QUESTION ID
S82-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. Adding π gives the opposite point (−a, −b) of (a, b), and tan = (−b)/(−a) = b/a, the same ratio.

GRADING NOTE Half a turn takes (a, b) to (−a, −b); both coordinates change sign, so the ratio is unchanged. The others use the wrong point, mistake π for a full turn, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

Use symmetry to evaluate cos(−2π/3).

A. 1/2, because cos is even: cos(−2π/3) = cos(π/3) = 1/2

B. −1/2, because cos is even: cos(−2π/3) = cos(2π/3) = −1/2

C. 1/2, because cos is odd: cos(−2π/3) = −cos(2π/3) = 1/2

D. −√3/2, because cos(−2π/3) = −sin(2π/3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

3052

QUESTION ID

S82-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1/2, because cos is even: cos(−2π/3) = cos(2π/3) = −1/2

GRADING NOTE cos(−x) = cos x. The others call cos odd, use sine instead of cosine, or use the reference angle π/3 with the wrong sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is f(x) = sin x cos x even, odd, or neither?

A. Even: a product of two functions is always even

B. Odd: f(−x) = sin(−x)cos(−x) = (−sin x)(cos x) = −f(x)

C. Even: f(−x) = sin(−x)cos(−x) = sin x cos x = f(x)

D. Neither: sin is odd but cos is even

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

3053

QUESTION ID

S82-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Odd: f(−x) = sin(−x)cos(−x) = (−sin x)(cos x) = −f(x)

GRADING NOTE Odd × even = odd. The others use sin(−x) = sin x, judge the product by its factors without computing f(−x), or use a false rule about products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 1

Explain the unit-circle definition • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which point on the unit circle represents angle x?

A. (cos x, x)

B. (sin x, cos x)

C. (x, sin x)

D. (cos x, sin x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

3055

QUESTION ID

S82-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. (cos x, sin x)

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In which direction from (1, 0) does a negative angle go on the unit circle? Answer Clockwise or Counterclockwise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

3056

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Positive angles go counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

tan x = sin x/cos x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

3057

QUESTION ID

S82-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The point for angle π is

A. (0, −1)

B. (−1, 0)

C. (0, 1)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

3058

QUESTION ID

S82-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 0)

GRADING NOTE Half turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

3059

QUESTION ID

S82-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 2

Even and odd functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Which is the definition of an even function?

A. $f(x + 2\pi) = f(x)$ for every x in the domain

B. $f(-x) = f(x)$ for every x in the domain

C. $f(-x) = -f(x)$ for every x in the domain

D. $f(x) > 0$ for every x in the domain

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

3061

QUESTION ID

S82-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. f(−x) = f(x) for every x in the domain

GRADING NOTE Symmetric about the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the definition of an odd function?

A. f(−x) = f(x) for every x in the domain

B. f(x) < 0 for every x in the domain

C. f(x + π) = −f(x) for every x in the domain

D. f(−x) = −f(x) for every x in the domain

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

3062

QUESTION ID

S82-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. f(−x) = −f(x) for every x in the domain

GRADING NOTE Symmetric about the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An odd function's graph is symmetric about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

3063

QUESTION ID

S82-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotation by 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

x^3 is

A. odd

B. neither

C. even

D. both

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q4/5

3064

QUESTION ID

S82-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. odd

GRADING NOTE $(-x)^3 = -x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the product of an odd function and an even function odd or even? Answer Odd or Even.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

3065

QUESTION ID

S82-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE $f(-x)g(-x) = (-f(x))g(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

3065

SET 82 • HSF-TF.A.4

BACKWARD 3

Reflect points over the axes • Grade 8 • Geometry • 8.G.A.3

Reflect (a, b) over the x-axis.

A. (b, a)

B. (−a, b)

C. (a, −b)

D. (−a, −b)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

3067

QUESTION ID

S82-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. (a, −b)

GRADING NOTE The y-coordinate changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

Rotate (a, b) by 180° about the origin.

A. (−a, −b)

B. (a, −b)

C. (−a, b)

D. (−b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

3068

QUESTION ID

S82-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (−a, −b)

GRADING NOTE Both coordinates change sign; (−b, a) is a 90° rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Reflecting over the x-axis keeps the y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

3069

QUESTION ID

S82-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a, b) → (a, −b): the x-coordinate stays and y changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

Reflecting (0.6, 0.8) over the x-axis gives

A. (0.8, 0.6)

B. (−0.6, 0.8)

C. (−0.6, −0.8)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

3070

QUESTION ID

S82-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (0.6, −0.8)

GRADING NOTE Negate y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

3070

ANSWER KEY Reduced copy of the question page, with its answer

Rotate the point (√3/2, 1/2) by 180° about the origin.

A. (√3/2, −1/2)

B. (−√3/2, −1/2)

C. (−1/2, √3/2)

D. (−√3/2, 1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

3071

QUESTION ID

S82-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. (−√3/2, −1/2)

GRADING NOTE Both coordinates change sign; the last is a 90° rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 1

Use periodicity and symmetry to evaluate and simplify • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(17π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

3073

QUESTION ID

S82-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Any equivalent form. 17π/6 − 2π = 5π/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find tan(−11π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

3074

QUESTION ID

S82-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. tan(−11π/4) = tan(−11π/4 + 3π) = tan(π/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(x + \pi) = -\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

3075

QUESTION ID

S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Opposite point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin(x + \pi) + \sin x$ equals

A. 1

B. $\sin 2x$

C. $2 \sin x$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

3076

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $\sin(x + \pi) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify cos(−x) + cos(x + 2π).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

3077

QUESTION ID

S82-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2cos x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 2

Restrict trigonometric functions so they have inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

How does periodicity prevent sin x from having an inverse on all real numbers?

A. Each value appears only once in every period, so periodicity changes nothing.

B. Periodicity makes sin increasing everywhere, so it has no turning points.

C. Each value repeats every 2π, so an output does not determine its input.

D. Periodic functions are undefined at multiples of 2π, which breaks the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

3079

QUESTION ID

S82-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Each value repeats every 2π, so an output does not determine its input.

GRADING NOTE Not one-to-one. The others are false: values repeat, sin is not increasing everywhere, and it is defined at multiples of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the restricted domain used for tan x, and why?

A. $-\pi/2 < x < \pi/2$: tan increases there and takes every real value once.

B. $0 < x < \pi$: tan is defined and increasing everywhere there.

C. $-\pi/2 \le x \le \pi/2$: tan is defined at both endpoints.

D. $0 \le x < 2\pi$: tan takes every real value there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

3080

QUESTION ID

S82-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $-\pi/2 < x < \pi/2$: tan increases there and takes every real value once.

GRADING NOTE tan is undefined at $\pm\pi/2$ and at $\pi/2$ inside $(0, \pi)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

3080

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

arccos gives values between 0 and π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

3081

QUESTION ID

S82-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arctan(−1) equals

A. 7π/4

B. 3π/4

C. −π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

3082

QUESTION ID

S82-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −π/4

GRADING NOTE In (−π/2, π/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is arcsin(−1/2) = −arcsin(1/2)?

A. arcsin(−1/2) = 7π/6 because sin(7π/6) = −1/2, which is −arcsin(1/2).

B. sin is even on [−π/2, π/2], so sin(−π/6) = 1/2, giving arcsin(−1/2) = π/6.

C. A minus sign can always be moved outside any function, so arcsin(−a) = −arcsin(a).

D. sin is odd on [−π/2, π/2], so sin(−π/6) = −1/2, giving arcsin(−1/2) = −π/6 = −arcsin(1/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

3083

QUESTION ID
S82-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. sin is odd on [−π/2, π/2], so sin(−π/6) = −1/2, giving arcsin(−1/2) = −π/6 = −arcsin(1/2).

GRADING NOTE Oddness on the restricted domain carries over to the inverse. The others use an output outside [−π/2, π/2], call sin even, or use a false rule about moving a minus sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSF-TF.B.6 • ADVANCED EXTENSION (+)

Choose and justify domain restrictions that make trigonometric functions invertible

MAIN	Algebra II main question	**HSF-TF.B.6**
BACKWARD 1	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**
BACKWARD 2	Exact values on the unit circle	**HSF-TF.A.2**
BACKWARD 3	Periodicity of trigonometric functions	**HSF-TF.A.4**
FORWARD 1	Compose trigonometric functions with their inverses	**HSF-TF.B.6**
FORWARD 2	Solve trigonometric equations with inverses	**HSF-TF.B.7**

SET 83 • HSF-TF.B.6

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Why does sin x have no inverse on all real numbers, and on which restricted domain does it have one?

A. sin repeats values, so it is not one-to-one; on $0 \le x \le \pi$ it is increasing and takes each value in $[-1, 1]$ once.

B. sin is not defined for all real numbers; on $0 \le x \le \pi$ it is defined and takes each value in $[-1, 1]$ once.

C. sin repeats values, so it is not one-to-one; on $-\pi/2 \le x \le \pi/2$ it is increasing and takes each value in $[-1, 1]$ once.

D. sin has outputs greater than 1 for large x; on $-\pi \le x \le \pi$ it takes each value in $[-1, 1]$ exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

3086

QUESTION ID
S83-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. sin repeats values, so it is not one-to-one; on $-\pi/2 \le x \le \pi/2$ it is increasing and takes each value in $[-1, 1]$ once.

GRADING NOTE On $[0, \pi]$ sin rises then falls; on $[-\pi, \pi]$ values repeat. The others use $[0, \pi]$ or $[-\pi, \pi]$, where values repeat, or claim false facts about the domain and outputs of sin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

What is the standard restricted domain of cos x, and why does it work?

A. $0 \le x \le 2\pi$: cos takes every value from −1 to 1 there, so it is enough.

B. $0 \le x \le \pi$: cos decreases from 1 to −1 there, taking each value once.

C. $-\pi/2 \le x \le \pi/2$: cos increases from −1 to 1 there, taking each value once.

D. $-\pi \le x \le 0$: cos decreases from 1 to −1 there, taking each value once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

3087

QUESTION ID

S83-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $0 \le x \le \pi$: cos decreases from 1 to −1 there, taking each value once.

GRADING NOTE On $[-\pi/2, \pi/2]$ cos rises and falls; on $[-\pi, 0]$ it increases. The others use $[-\pi/2, \pi/2]$, where cos rises then falls, $[0, 2\pi]$, where values repeat, or $[-\pi, 0]$, where cos increases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

What is the standard restricted domain of tan x, and why does it work?

A. $-\pi/2 < x < \pi/2$: tan increases from $-\infty$ to ∞ there, taking each real value once.

B. $0 < x < \pi$: tan is defined and increasing at every point of that interval.

C. $0 \le x < \pi/2$: tan increases there and takes each real value exactly once.

D. $-\pi/2 \le x \le \pi/2$: tan increases there, including its values at both endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

3088

QUESTION ID

S83-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-\pi/2 < x < \pi/2$: tan increases from $-\infty$ to ∞ there, taking each real value once.

GRADING NOTE tan is undefined at $\pm\pi/2$ and takes only nonnegative values on $[0, \pi/2)$. The others include $\pi/2$ (where tan is undefined), include the endpoints $\pm\pi/2$, or use $[0, \pi/2)$, which gives only nonnegative values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is arcsin(1/2), and why is it not 5π/6?

A. π/6 and 5π/6: arcsin gives every angle in [0, 2π) whose sine is 1/2.

B. π/3: arcsin(1/2) is the angle whose cosine is 1/2, not 5π/6.

C. 5π/6: arcsin gives the largest angle in [0, 2π) whose sine is 1/2.

D. π/6: arcsin outputs only angles in [−π/2, π/2], and 5π/6 is outside that interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

3089

QUESTION ID

S83-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. π/6: arcsin outputs only angles in [−π/2, π/2], and 5π/6 is outside that interval.

GRADING NOTE An inverse function has one output. The others give an angle outside the range of arcsin, give two outputs for one input, or use the cosine value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arccos(−1/2) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

3090

QUESTION ID
S83-M-Q5

TYPE
Short answer

CORRECT ANSWER
2π/3

GRADING NOTE Form required: exact value in [0, π].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 1

Restrict a domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

On which domain is $f(x) = x^2$ invertible?

A. all real numbers

B. $x \geq -2$

C. $x \geq 0$

D. $-1 \leq x \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

3092

QUESTION ID

S83-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. $x \geq 0$

GRADING NOTE f is one-to-one on one side of the vertex only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

3092

ANSWER KEY Reduced copy of the question page, with its answer

Why must a function be strictly increasing or strictly decreasing on an interval to be invertible there?

A. Then the function is defined at every real number, inside and outside the interval.

B. Then the function has no zeros on the interval, so it never crosses the x-axis.

C. Then the outputs are all positive on the interval, so each can be undone.

D. Then no output repeats on the interval, so each output comes from exactly one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

3093

QUESTION ID

S83-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Then no output repeats on the interval, so each output comes from exactly one input.

GRADING NOTE Strict monotonicity on an interval implies one-to-one there. The others name properties (domain, zeros, positive outputs) that do not make a function one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

True or false?

A function that is always increasing on an interval is one-to-one there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

3094

QUESTION ID

S83-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Monotonic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 1)^2$ is one-to-one on

A. $-1 \le x \le 3$

B. all reals

C. $x \ge 1$

D. $x \ge 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

3095

QUESTION ID

S83-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x \ge 1$

GRADING NOTE One side of the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of $f(x) = x^2$ on $x \geq 0$?

A. $-\sqrt{x}$

B. $\sqrt{x}$

C. $x/2$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

3096

QUESTION ID

S83-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. √x

GRADING NOTE The outputs of the inverse must be ≥ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 2

Exact values on the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(π/6) and sin(5π/6) exactly.

sin(π/6) = _____

sin(5π/6) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

3098

QUESTION ID

S83-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/2; 1/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3) and cos(4π/3) exactly.

cos(2π/3) = _____

cos(4π/3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

3099

QUESTION ID

S83-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2; −1/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

True or false?

$\tan(\pi/4) = \tan(3\pi/4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

3100

QUESTION ID

S83-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\tan(\pi/4) = 1$ and $\tan(3\pi/4) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

3100

Which angle in [0, π] has cos = −1/2?

A. 2π/3

B. π/3

C. 4π/3

D. 5π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

3101

QUESTION ID

S83-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π/3

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give $\sin(-\pi/6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

3102

QUESTION ID

S83-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-1/2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-B2-Q5**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 3

Periodicity of trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

3104

QUESTION ID

S83-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of tan x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

3105

QUESTION ID

S83-B3-Q2

TYPE

Short answer

CORRECT ANSWER

π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

True or false?

A periodic function cannot be one-to-one on its whole domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

3106

QUESTION ID
S83-B3-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Values repeat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin x = 0 for

A. x = π/2 + kπ

B. no x

C. x = 2kπ only

D. x = kπ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

3107

QUESTION ID

S83-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. x = kπ

GRADING NOTE Every multiple of π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which two angles both have cosine 1?

A. 0 and 2π

B. 0 and π

C. π and 2π

D. π/2 and 3π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

3108

QUESTION ID

S83-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 0 and 2π

GRADING NOTE cos x = 1 at multiples of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 1

Compose trigonometric functions with their inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(sin(5π/6)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

3110

QUESTION ID

S83-F1-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate sin(arcsin(0.3)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

3111

QUESTION ID

S83-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

arccos(cos x) = x for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

3112

QUESTION ID

S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only for 0 ≤ x ≤ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arctan(tan(3π/4)) equals

A. −π/4

B. −3π/4

C. 3π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

3113

QUESTION ID

S83-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −π/4

GRADING NOTE Into (−π/2, π/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the domain and range of arccos?

A. all real numbers; range [0, π]

B. domain [0, π]; range [−1, 1]

C. domain [−1, 1]; range [−π/2, π/2]

D. domain [−1, 1]; range [0, π]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

3114

QUESTION ID

S83-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. domain [−1, 1]; range [0, π]

GRADING NOTE arccos inverts cos on [0, π].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 2

Solve trigonometric equations with inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sin x = 0.4$ on $0 \le x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

3116

QUESTION ID

S83-F2-Q1

TYPE

Short answer

CORRECT ANSWER

{0.41, 2.73}

GRADING NOTE Form required: both. arcsin(0.4) and π − arcsin(0.4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve cos x = 0.4 on 0 ≤ x < 2π. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

3117

QUESTION ID

S83-F2-Q2

TYPE

Short answer

CORRECT ANSWER

{1.16, 5.12}

GRADING NOTE Form required: both. arccos(0.4) and 2π − arccos(0.4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

True or false?

For $0 < c < 1$, arcsin c gives only one of the two solutions of $\sin x = c$ on $[0, 2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

3118

QUESTION ID

S83-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The other is π − arcsin c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The second solution of $\sin x = c$ in $[0, 2\pi)$ (with $0 < c < 1$) is

A. $\pi + \arcsin c$

B. $-\arcsin c$

C. $\pi - \arcsin c$

D. $2\pi - \arcsin c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

3119

QUESTION ID

S83-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π − arcsin c

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For $-1 < c < 1$, how is the restricted range of arccos used to find both solutions of $\cos x = c$ on $[0, 2\pi)$?

A. arccos c gives the solution in (0, π); the other is 2π − arccos c, by symmetry over the x-axis.

B. arccos c gives both solutions at once, because its range is the whole interval [0, 2π).

C. arccos c gives the solution in (0, π); the other is π − arccos c, by symmetry over the y-axis.

D. arccos c gives the solution in (−π/2, π/2); the other is arccos c + π, by periodicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

3120

QUESTION ID

S83-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. arccos c gives the solution in (0, π); the other is 2π − arccos c, by symmetry over the x-axis.

GRADING NOTE cos is even: cos(2π − a) = cos(−a) = cos a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSF-TF.B.7 • ADVANCED EXTENSION (+)

Solve trigonometric equations in models with inverse functions

MAIN	Algebra II main question	**HSF-TF.B.7**
BACKWARD 1	Evaluate inverse trigonometric functions on restricted domains	**HSF-TF.B.6**
BACKWARD 2	Parameters of sinusoidal models	**HSF-TF.B.5**
BACKWARD 3	Symmetry of unit-circle solutions	**HSF-TF.A.2**
FORWARD 1	Find all real solutions and solve equations that factor	**HSF-TF.B.7**
FORWARD 2	Solve equations with sum and double-angle identities	**HSF-TF.C.9**

SET 84 • HSF-TF.B.7

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\sin x - 1 = 0$ for $0 \le x < 2\pi$. Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

3123

QUESTION ID

S84-M-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3\cos x = -2$ for $0 \le x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

3124

QUESTION ID

S84-M-Q2

TYPE

Short answer

CORRECT ANSWER

{2.30, 3.98}

GRADING NOTE Form required: both. arccos(−2/3) and 2π − arccos(−2/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m after t seconds. Find every time in $0 \le t < 30$ when $h = 17$ m, in seconds, separated by commas.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

3125

QUESTION ID
S84-M-Q3

TYPE
Short answer

CORRECT ANSWER
{2.5, 12.5}

GRADING NOTE Form required: both. $\sin(\pi t/15) = 1/2$ gives $\pi t/15 = \pi/6$ or $5\pi/6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth is d(t) = 5 + 4cos(πt/6) ft after t hours. Find every time in 0 ≤ t < 12 when d = 7 ft, in hours, separated by commas.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

3126

QUESTION ID

S84-M-Q4

TYPE

Short answer

CORRECT ANSWER

{2, 10}

GRADING NOTE Form required: both. cos(πt/6) = 1/2 gives πt/6 = π/3 or 5π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\tan x = 3$ for $0 \le x < 2\pi$. Give both solutions to the nearest hundredth, separated by a comma. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

3127

QUESTION ID

S84-M-Q5

TYPE

Short answer

CORRECT ANSWER

{1.25, 4.39}

GRADING NOTE Form required: both. arctan 3 and arctan 3 + π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 1

Evaluate inverse trigonometric functions on restricted domains • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

3129

QUESTION ID

S84-B1-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arccos(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

3130

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-B1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

True or false?

arctan 3 is between 0 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

3131

QUESTION ID

S84-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Positive input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arccos(−2/3) is in quadrant

A. II

B. III

C. I

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

3132

QUESTION ID

S84-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. II

GRADING NOTE Range [0, π], negative cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arctan 3 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

3133

QUESTION ID

S84-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1.25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-B1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 2

Parameters of sinusoidal models • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In h(t) = 12 + 10sin(πt/15), what is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

3135

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rider's height is h(t) = 12 + 10sin(πt/15) m. Setting h(t) = 17 gives an equation sin(πt/15) = c. Find c.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

3136

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 10sin(πt/15) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

True or false?

d(t) = 5 + 4cos(πt/6) has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q3/5

3137

QUESTION ID

S84-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The maximum is 5 + 4 = 9; 5 is the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q3/5

Isolating the cosine in 5 + 4cos(πt/6) = 7 gives

A. cos(πt/6) = 1/2

B. cos(πt/6) = 3

C. cos(πt/6) = 2

D. cos(πt/6) = 7/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

3138

QUESTION ID

S84-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos(πt/6) = 1/2

GRADING NOTE Subtract 5, divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the midline of d(t)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

3139

QUESTION ID

S84-B2-Q5

TYPE

Short answer

CORRECT ANSWER

d = 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 3

Symmetry of unit-circle solutions • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with sin = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

3141

QUESTION ID

S84-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with cos = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

3142

QUESTION ID

S84-B3-Q2

TYPE

Short answer

CORRECT ANSWER

π/3 and 5π/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

True or false?

tan x = c has two solutions in [0, 2π) that differ by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

3143

QUESTION ID

S84-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Period π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $-1 < c < 1$, $\cos x = c$ has a solution a in $(0, \pi)$. The other solution in $[0, 2\pi)$ is

A. $2\pi - a$

B. $\pi + a$

C. $-a$

D. $\pi - a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

3144

QUESTION ID

S84-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $2\pi - a$

GRADING NOTE Reflection over the x-axis; $-a$ is not in $[0, 2\pi)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with tan = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

3145

QUESTION ID

S84-B3-Q5

TYPE

Short answer

CORRECT ANSWER

π/4 and 5π/4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 1

Find all real solutions and solve equations that factor • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Which gives all real solutions of $2\sin x - 1 = 0$?

A. $x = \pi/3 + 2k\pi$ or $x = 2\pi/3 + 2k\pi$, k an integer

B. $x = \pi/6 + k\pi$, k an integer

C. $x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$, k an integer

D. $x = \pi/6 + 2k\pi$, k an integer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

3147

QUESTION ID

S84-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. $x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$, k an integer

GRADING NOTE sin x = 1/2 at π/6 and 5π/6 in each period. π/6 + π gives sin = −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

Solve $2\sin^2 x - \sin x = 0$ for $0 \le x < 2\pi$. List all solutions exactly, separated by commas.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

3148

QUESTION ID

S84-F1-Q2

TYPE

Short answer

CORRECT ANSWER

{0, π/6, 5π/6, π}

GRADING NOTE Form required: all four. $\sin x(2\sin x - 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

True or false?

cos x = 2 has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

3149

QUESTION ID

S84-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many solutions does $\sin(2x) = 1/2$ have in $0 \le x < 2\pi$?

A. 2

B. 4

C. 8

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

3150

QUESTION ID

S84-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 2x runs over two periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\cos^2 x = 1/4$ for $0 \le x < 2\pi$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

3151

QUESTION ID

S84-F1-Q5

TYPE

Short answer

CORRECT ANSWER

π/3, 2π/3, 4π/3, 5π/3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 2

Solve equations with sum and double-angle identities • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Use sin 2x = 2sin x cos x to solve sin 2x = cos x for $0 \le x < 2\pi$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

3153

QUESTION ID

S84-F2-Q1

TYPE

Short answer

CORRECT ANSWER

π/6, π/2, 5π/6, 3π/2

GRADING NOTE Form required: all four. cos x(2sin x − 1) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\cos 2x = 1 - 2\sin^2 x$ to solve $\cos 2x = \sin x$ for $0 \le x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

3154

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

π/6, 5π/6, 3π/2

GRADING NOTE Form required: all three. $2\sin^2 x + \sin x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

True or false?

$\sin(x + \pi/2) = \cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

3155

QUESTION ID

S84-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sum formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos 2x equals

A. $1 + 2\sin^2 x$

B. $\cos^2 x - \sin^2 x$

C. $\cos^2 x + \sin^2 x$

D. $2\cos x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

3156

QUESTION ID
S84-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $\cos^2 x - \sin^2 x$

GRADING NOTE Double angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

Why are identities useful for solving equations that contain different trigonometric functions?

A. They rewrite the equation in one function (or let it factor), so inverse functions can be applied.

B. They turn every equation into a linear equation that has exactly one solution.

C. They remove the need to check the solutions in the original equation.

D. They change the period of each function to 2π, so the graphs match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5

3157

QUESTION ID

S84-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. They rewrite the equation in one function (or let it factor), so inverse functions can be applied.

GRADING NOTE For example sin 2x = sin x becomes sin x(2cos x − 1) = 0. The others are false: identities do not make equations linear, do not remove the need to check, and do not change periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5